

Generalized Poincaré inequality for quantum Markov semigroups

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Abstract

We prove a noncommutative $\mathbf{p}$ -Poincaré inequality for GNS-detailed-balance quantum Markov semigroups (QMSs) on non-tracial σ -finite von Neumann algebras, assuming only the existence of a spectral gap. Extending the semi-commutative results of Huang and Tropp [21], we first establish the inequality for tracial von Neumann algebras. We use Markov dilations to obtain chain-rule estimates for Dirichlet forms and amalgamated free products to define an appropriate noncommutative derivation. We then extend the result to QMSs satisfying GNS detailed balance on non-tracial σ -finite von Neumann algebras, using Haagerup's reduction and Kosaki's interpolation theorem. As applications, we recover sub-exponential concentration inequalities and estimate the Lipschitz and completely bounded Lipschitz diameters of the QMSs.

Keywords: Poincaré inequalities, quantum Markov semigroups, Markov dilations, amalgamated free products, Haagerup's L^p spaces, Kosaki's L^p interpolation, Haagerup reduction, concentration inequality, Lipschitz diameter, completely bounded Lipschitz diameter.

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1 Introduction

L^p Poincaré inequalities have a long history in harmonic analysis, geometry, and probability. They control higher moments and yield concentration results. Building on Lust-Piquard's dimension-free estimates for discrete Riesz transforms [33, 34], obtained using noncommutative techniques by Pisier [40], Ben Efraim and Lust-Piquard [6] proved dimension-free L^p Poincaré inequalities for the discrete cube and corresponding inequalities for the CAR algebra. These arguments adapt Gaussian methods of Maurey and Pisier to special structures, such as Bernoulli and fermionic settings. Similarly, Junge and Zeng [26] established broader noncommutative results that rely on a positive curvature condition, which is stronger than a spectral gap. Huang and Tropp [21] later obtained semi-commutative matrix-valued L^p moment bounds from a spectral gap alone. In this work, we extend this spectral-gap principle from the semi-commutative setting to fully noncommutative quantum Markov semigroups satisfying GNS detailed balance on non-tracial σ -finite von Neumann algebras.

We start with the Poincaré spectral-gap estimate [43]. In its classical Euclidean form, the Poincaré inequality states that if $\Omega \subseteq \mathbb{R}^d$ is a bounded, connected open set with Lipschitz boundary, then there exists a constant $\alpha > 0$, depending only on Ω , such that every smooth compactly supported function f on Ω satisfies

$$\alpha \int_{\Omega} |f(x) - f_{\Omega}|^2 dx \leq \int_{\Omega} \|\nabla f(x)\|_2^2 dx,$$

where $f_{\Omega} = \frac{1}{|\Omega|} \int_{\Omega} f(x) dx$ denotes the mean of f .

Meyer [36] reformulated the classical gradient in the semigroup setting as the *carré du champ*, which we refer to as the gradient form. This framework was subsequently developed systematically by Bakry and Émery [5]. Let $(T_t)_{t \geq 0} = e^{-tL}$ be a conservative, μ -symmetric Markov semigroup on a probability space (X, μ) , and let Γ be its gradient form. The semigroup is said to have spectral gap $\alpha > 0$ if

$$\|f - \mu(f)\|_{L^2(X, \mu)} \leq \frac{1}{\sqrt{\alpha}} \|\Gamma(f, f)\|_{L^1(X, \mu)}^{1/2}. \quad (1)$$

More generally, for $p, q/2 \in [1, \infty]$, we say that T_t satisfies the *Poincaré inequality* $\text{PI}(p, q)$ with constant $C > 0$ if

$$\|f - \mu(f)\|_{L^p(X, \mu)} \leq C \|\Gamma(f, f)\|_{L^{q/2}(X, \mu)}^{1/2}. \quad (2)$$

We use the same terminology in the noncommutative setting. Let $(\mathcal{M}, \tau)$ be a tracial von Neumann algebra with faithful normal tracial state τ , and let $(T_t)_{t \geq 0} = e^{-tL}$ be a weak*-continuous, τ -symmetric quantum Markov semigroup. Its gradient form is

$$\Gamma(x, y) = \frac{1}{2} (L(x^*)y + x^*L(y) - L(x^*y)), \quad x, y \in \text{Dom}(L).$$

The fixed-point space

$$\mathcal{N} = \{x \in \mathcal{M} : T_t(x) = x \text{ for all } t \geq 0\}$$

is a von Neumann subalgebra. We denote by $E : \mathcal{M} \rightarrow \mathcal{N}$ the canonical τ -preserving conditional expectation.

We have the following noncommutative analogue of eq. (2).

Definition 1.1 (Tracial Poincaré inequality $\text{PI}(p, q)$). *For $p, q/2 \in [1, \infty]$, the QMS satisfies the Poincaré inequality $\text{PI}(p, q)$ with constant $C > 0$ if, for every self-adjoint $x \in \text{Dom}(L)$,*

$$\|x - E(x)\|_{L^p(\mathcal{M}, \tau)} \leq C \|\Gamma(x, x)\|_{L^{q/2}(\mathcal{M}, \tau)}^{1/2}. \quad (3)$$

In particular, the spectral gap is exactly $\text{PI}(2, 2)$ (with $C = \alpha^{-1/2}$).

With the above definition, the tracial form of our main result is the following:

Theorem 1.2 ($\text{PI}(p, p)$). *Let $(\mathcal{M}, \tau)$ be a tracial von Neumann algebra and $(T_t)_{t \geq 0} = e^{-tL}$ a τ -symmetric quantum Markov semigroup with fixed-point algebra $\mathcal{N}$ and conditional expectation $E : \mathcal{M} \rightarrow \mathcal{N}$. If $\{T_t\}_{t \geq 0}$ has a spectral gap $\alpha > 0$, then for every self-adjoint x in the domain of L and $p = 2$ or $p \geq 3$, it satisfies the Poincaré inequality $\text{PI}(p, p)$ with constant $\frac{p}{\sqrt{2\alpha}}$, i.e.,*

$$\|x - E(x)\|_{L^p(\mathcal{M}, \tau)} \leq \frac{p}{\sqrt{2\alpha}} \|\Gamma(x, x)\|_{L^{p/2}(\mathcal{M}, \tau)}^{1/2}.$$

The proof uses the argument of Huang and Tropp as a stepping stone. In particular, they established a chain rule estimate relating the Dirichlet form to the associated gradient form. A further ingredient in their argument is a symmetrization procedure, based on doubling variables in the product space of a Markov process, which leads to key cancellations. Our proof, in addition, identifies the appropriate dilation framework for quantum Markov semigroups, which allows us to derive chain rule estimates directly in the noncommutative setting. Since there are obstacles to generalizing their product measure symmetrization procedure, we instead formulate it in terms of amalgamated free products, providing a natural setting for noncommutative algebras.

Beyond the tracial setting: GNS-detailed balance via Haagerup reduction.

Many physically relevant QMS are not tracial but are reversible with respect to a faithful normal state ϕ (GNS-symmetry/detailed balance). In that case, one works on Haagerup L^p spaces [18] $L^p(\mathcal{M})$ and uses the GNS inner product in Section 4.1.

Our strategy reduces the non-tracial problem to the tracial one via Haagerup's reduction (crossed-product) technique. We embed $(\mathcal{M}, \phi)$ into a semifinite crossed product $(\widehat{\mathcal{M}}, \widehat{\phi})$, which admits an increasing sequence of tracial subalgebras $(\mathcal{M}_k, \tau_k)$. The semigroup is lifted to be $\widehat{\phi}$ -detailed balance on $\widehat{\mathcal{M}}$ and τ_k -symmetric on $\mathcal{M}_k$, where a $\text{PI}(p, q)$ inequality on $\widehat{\mathcal{M}}$ is obtained by tracial $\text{PI}(p, p)$ on each slice $(\mathcal{M}_k, \tau_k)$. The

result is then descended to $\mathcal{M}$ via conditional expectations compatible with the modular action, giving a GNS- ϕ -detailed balance version of $\text{PI}(p, p)$. Throughout the paper, we abbreviate the associated Haagerup gradient seminorm by

$$\|a\|_{\Gamma_p} := \|\Gamma_p(a, a)\|_{L^{p/2}(\mathcal{M})}^{1/2}.$$

Thus the inequality takes the form

$$\|a - E_p^\phi(a)\|_{L^p(\mathcal{M})} \leq C\|a\|_{\Gamma_p},$$

where E_p^ϕ denotes the L^p extension of the ϕ -preserving conditional expectation onto the fixed-point algebra, Γ_p the L^p version of the gradient form, and $\|\cdot\|_{L^p(\mathcal{M})}$ the norm in Haagerup's L^p space. A key advantage of Haagerup's L^p construction is its functoriality: every normal $*$ -homomorphism between von Neumann algebras induces canonically bounded maps on the associated L^p -spaces, compatible with composition, duality, and interpolation. Our main result is stated in this framework as follows:

Theorem 1.3 (cf Theorem 4.17 and Theorem 4.18). *Let $(\mathcal{M}, \phi)$ be a σ -finite von Neumann algebra with faithful normal state ϕ , and let $(T_t)_{t \geq 0}$ be a GNS- ϕ -detailed-balance QMS with fixed-point algebra $\mathcal{N}$. Suppose (T_t) has an $L^2(\mathcal{M}, \phi)$ spectral gap $\alpha > 0$. Let $E_p^\phi : L^p(\mathcal{M}) \rightarrow L^p(\mathcal{N})$ be the induced conditional expectation. Then for every $p = 2$ and every $p \geq 3$, and for all self-adjoint $a \in L^p(\mathcal{M})_{\text{sa}}$ in the domain, we have*

$$\|a - E_p^\phi(a)\|_{L^p(\mathcal{M})} \leq \frac{p}{\sqrt{2\alpha}} \|a\|_{\Gamma_p}.$$

Furthermore, for general $a \in L^p(\mathcal{M})$ in the domain,

$$\|a - E_p^\phi(a)\|_{L^p(\mathcal{M})} \leq \frac{p}{\sqrt{2\alpha}} (\|a\|_{\Gamma_p} + \|a^*\|_{\Gamma_p}).$$

Sub-exponential concentration inequalities.

Poincaré-type inequalities are closely connected to the concentration of measure. In the semi-commutative setting, such a principle was established by Huang and Tropp [21, Thm 2.7] for Markov processes with a spectral gap, showing sub-exponential concentration. Their approach relies on matrix inequalities and dilation techniques, and improves upon earlier results of Aoun, Banna, and Youssef [1]. In particular, Huang and Tropp introduce a symmetrization (or doubling-of-variables) argument that avoids technical difficulties present in the earlier approach.

We show that the $\text{PI}(p, p)$ inequality proved in this work implies a *sub-exponential* concentration inequality for observables measured in the Lipschitz seminorm induced by the gradient form as below.

Corollary (cf Corollary 5.7). *Let $T_t = e^{-tL}$ be a GNS- ϕ -detailed balance QMS with spectral gap $\alpha > 0$ and ϕ -preserving conditional expectation E^ϕ onto its fixed-point*

algebra. Then for any $x \in \mathcal{M}$ and large $t > 0$,

$$\mathbb{P}_\phi(|x - E^\phi(x)| > t) \leq 2 \exp\left(-\frac{\sqrt{\alpha}t}{8\sqrt{2}e\|x\|_{\text{Lip}_\Gamma}}\right).$$

Here the Lipschitz seminorm is defined by

$$\|x\|_{\text{Lip}_\Gamma} = \max\left\{\|\Gamma(x, x)\|_\infty^{1/2}, \|\Gamma(x^*, x^*)\|_\infty^{1/2}\right\},$$

to be distinguished from the Lipschitz complexity defined in [2, eq. (1.1)].

This reveals a qualitative distinction between spectral-gap and stronger functional-inequality regimes. An L^2 spectral gap alone does not yield Gaussian or Talagrand-type concentration in general, while one can still obtain sub-exponential tails. We compare this regime with the stronger concentration implied by modified logarithmic Sobolev inequalities (MLSI) and discuss a birth-death process Example 5.9 that rules out Gaussian decay under a uniform spectral gap assumption alone.

For finite-dimensional matrix algebras, we establish estimates for the Lipschitz diameters and completely bounded Lipschitz diameters Corollary 5.10. We summarize the main results and local implications in Figure 1.

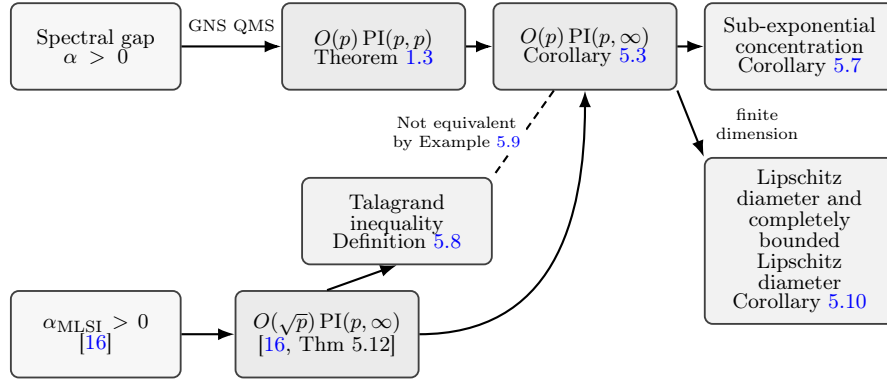


Fig. 1 Roadmap of the logical structure of the main results. Solid arrows denote implications, while dashed lines indicate that no implication is asserted. The birth-death example shows a violation of the dimension-free geometric Talagrand inequality, but the $O(p)$ $\text{PI}(p, p)$ estimate holds.

Related work.

To our knowledge, this is the first result establishing a noncommutative (p, p) -type Poincaré inequality under only an L^2 spectral gap assumption, within the class of QMSs satisfying detailed balance. Earlier works obtain Poincaré inequalities with different index pairs and under stronger functional or curvature assumptions. In the

noncommutative setting, such curvature bounds are often difficult to verify beyond specific model classes [38, 51].

In [16], assuming a modified logarithmic Sobolev inequality (MLSI) with constant $\alpha_{MLSI} > 0$, the authors prove $\text{PI}(p, \infty)$ with optimal order $O(\sqrt{p})$. The relation between our $\text{PI}(p, p)$ and the MLSI-based $\text{PI}(p, \infty)$ regime is not clear, and neither inequality evidently forces the other. MLSI further implies the geometric Talagrand inequality from [15, Remark 6.9], while we illustrate in Example 5.9 that Talagrand-type bounds do not hold for all QMS with only an L^2 spectral gap.

Under the noncommutative Γ_2 -condition

$$\tau(\Gamma(T_t x, T_t x)y) \leq C e^{-2\alpha t} \tau(T_t \Gamma(x, x)y) \quad (x \in \text{Dom}(A^{1/2}), y \in \mathcal{N}_+),$$

[26] obtains both (p, p) - and (p, ∞) -type inequalities with constants of order $O(p)$ and $O(\sqrt{p})$, respectively. Since both MLSI and Γ_2 imply an L^2 spectral gap, while the converse fails in general, our hypothesis is strictly weaker.

As for applications, [13] uses a $(2, \infty)$ -type Poincaré inequality to derive lower bounds on quantum circuit depth. Using our framework, we say that a $(2, \infty)$ -type Poincaré inequality holds for a GNS- ϕ -detailed-balance QMS if

$$\|x - E^\phi(x)\|_{L^2(\mathcal{M}, \phi)_L} \leq C \|\Gamma(x, x)\|_\infty^{1/2}, \quad x = x^*,$$

where $\|x\|_{L^2(\mathcal{M}, \phi)_L} = \sqrt{\tau(D_\phi x^* x)}$. We show $(2, \infty)$ -type Poincaré inequality in Corollary 5.3 by choosing $p = 2, a = x D_\phi^{1/2}$, where D_ϕ is the density of ϕ . The relationship between the $(2, \infty)$ -type Poincaré inequality used in [13] and our formulation eq. (3) (up to a dimension factor) is made explicit in [2, (3.9), (4.16)]. Further connections to circuit complexity appear in [3, 10, 14, 32]. When detailed balance (or symmetry) fails, $\text{PI}(p, q)$ generally breaks down; instead, space-time Poincaré inequalities under weaker symmetry assumptions are established in [31]. We also mention related unpublished work of Li Gao, Zhendong Xu, and Hao Zhang, communicated to the first-named author, which provides an independent proof for the tracial case.

Organization of this paper.

Section 2 collects notation and background on von Neumann algebras, quantum Markov semigroups, Markov dilations, and amalgamated free products. Section 3 introduces the swap operator and the amalgamated free product derivation, proves the generalized Klein inequality used in the argument, and establishes Theorem 1.2. Section 4 reviews Haagerup and Kosaki L^p spaces, states the GNS detailed balance framework, and proves Theorem 1.3 via Haagerup reduction. Section 5 presents examples and applications, including sub-exponential concentration and finite-dimensional estimates for the Lipschitz diameter and its completely bounded analogue. The appendices give the Markov-dilation chain estimate, spectral-gap stability under regularization, domain convergence, and the convexification argument needed in the general case.

Notation. We use script letters $\mathcal{M}, \mathcal{N}$ for von Neumann algebras, τ for a trace and ϕ for a non-tracial state. The identity operator is denoted by 1 and the identity map by Id . For $x \in \mathcal{M}$, x^* denotes the adjoint and Φ_* the pre-adjoint of a normal map Φ . Let $(T_t)_{t \geq 0} = e^{-tL}$ be a QMS with generator L , and denote by Γ and $\mathcal{E}$ the associated gradient and Dirichlet forms.

2 Preliminaries

2.1 Von Neumann algebras and conditional expectations

Let $(\mathcal{M}, \phi)$ denote a von Neumann algebra $\mathcal{M}$ equipped with a normal, faithful state ϕ . Write $(\pi_\phi, H_\phi, \xi_\phi)$ for the GNS (Gelfand–Naimark–Segal) representation of $\mathcal{M}$ with respect to ϕ , where $\pi_\phi : \mathcal{M} \rightarrow B(H_\phi)$ is a faithful normal $*$ -representation and $\xi_\phi \in H_\phi$ is cyclic and separating for $\pi_\phi(\mathcal{M})$, satisfying

$$\phi(x) = \langle \xi_\phi, \pi_\phi(x) \xi_\phi \rangle, \quad x \in \mathcal{M}.$$

Here ξ_ϕ is *cyclic* for $\mathcal{M}$ if $\overline{\mathcal{M}\xi_\phi} = H_\phi$, and *separating* if $x\xi_\phi = 0$ implies $x = 0$, or equivalently $\overline{\mathcal{M}'\xi_\phi} = H_\phi$. After identifying $\mathcal{M}$ with $\pi_\phi(\mathcal{M})$, we may regard $\mathcal{M}$ as a von Neumann algebra on H_ϕ with cyclic and separating vector ξ_ϕ .

Define the densely defined antilinear map

$$S : H_\phi \rightarrow H_\phi, \quad S(x\xi_\phi) = x^*\xi_\phi, \quad x \in \mathcal{M}.$$

By Tomita–Takesaki theory [47], the closure $\overline{S}$ admits the polar decomposition $\overline{S} = J\Delta^{1/2}$, where J is a conjugate-linear unitary and Δ is a positive self-adjoint operator. We call J the *modular conjugation* and $\Delta := \overline{S}^* \overline{S}$ the *modular operator* associated with ϕ . The modular automorphism group $(\sigma_t^\phi)_{t \in \mathbb{R}}$ is then given by

$$\sigma_t^\phi(x) = \Delta^{it} x \Delta^{-it}, \quad x \in \mathcal{M}, t \in \mathbb{R}. \quad (4)$$

Let $\mathcal{N} \subset \mathcal{M}$ be a von Neumann subalgebra. If $\mathcal{N}$ is globally invariant under the modular automorphism group,

$$\sigma_t^\phi(\mathcal{N}) = \mathcal{N} \quad (t \in \mathbb{R}),$$

then by Takesaki's theorem [45, 47] there exists a *unique* ϕ -preserving conditional expectation

$$E_\phi : \mathcal{M} \rightarrow \mathcal{N}$$

such that

- (i) $E_\phi(y) = y$ for all $y \in \mathcal{N}$;
- (ii) $E_\phi(y_1 x y_2) = y_1 E_\phi(x) y_2$ for all $x \in \mathcal{M}$, $y_1, y_2 \in \mathcal{N}$;
- (iii) $\phi(E_\phi(x)) = \phi(x)$ for all $x \in \mathcal{M}$.

It follows that E_ϕ is normal, completely positive, contractive, and $\mathcal{N}$ -bimodular. For more background on von Neumann algebras, see [46, 47]. In the remainder of this section, we discuss L^p spaces in the tracial case, postponing the non-tracial case to Section 4.1.

Let $(\mathcal{M}, \tau)$ be a von Neumann algebra equipped with a faithful normal tracial state τ . In this setting, one can define noncommutative L^p spaces in complete analogy with the classical case. For $1 \leq p < \infty$, the L^p norm is

$$\|x\|_p := \tau(|x|^p)^{1/p}, \quad |x| = (x^*x)^{1/2}, \quad x \in \mathcal{M},$$

and $\|x\|_\infty := \|x\|$ is the operator norm. The completion of $\mathcal{M}$ under this norm gives the noncommutative L^p space $L^p(\mathcal{M}, \tau)$.

We write $\mathcal{M}_{\text{sa}}$ for the self-adjoint part of $\mathcal{M}$ and $\mathcal{M}_+$ for the positive cone. The case $p = 2$ is of particular importance: $L^2(\mathcal{M}, \tau)$ carries the inner product

$$\langle x, y \rangle_{L^2(\mathcal{M})} := \tau(x^*y), \quad x, y \in \mathcal{M},$$

under which $\mathcal{M}$ embeds densely and isometrically. This is exactly the GNS Hilbert space H_ϕ associated with τ .

2.2 Functional inequalities

In this section, we introduce the Poincaré inequality for trace-symmetric quantum Markov semigroups. Let $(T_t)_{t \geq 0} : \mathcal{M} \rightarrow \mathcal{M}$ be a quantum Markov semigroup (QMS) on $\mathcal{M}$, i.e., a family of normal, unital, completely positive maps (UCP maps) satisfying

- i) $T_0 = \text{Id}_\mathcal{M}$ and $T_{t+s} = T_t \circ T_s$ for all $s, t \geq 0$,
- ii) for each $x \in \mathcal{M}$, the map $t \mapsto T_t(x)$ is weak*-continuous.

We write formally $T_t = e^{-tL}$, where the semigroup generator (Lindbladian) L is given by

$$L(x) := \text{w}^*\text{-}\lim_{t \rightarrow 0} \frac{1}{t}(x - T_t(x)), \quad (5)$$

on the domain of those $x \in \mathcal{M}$ for which the limit exists. Assume now that $\mathcal{M}$ is finite and is equipped with a faithful normal tracial state τ ; we write this pair as $(\mathcal{M}, \tau)$. We define a QMS to be τ -symmetric if it is self-adjoint with respect to the trace, i.e.,

$$\tau(x^*T_t(y)) = \tau(T_t(x)^*y), \quad x, y \in \mathcal{M}, t \geq 0. \quad (6)$$

This definition implies that L is positive and τ -symmetric on $L^2(\mathcal{M}, \tau)$. Its fixed-point algebra

$$\mathcal{N} := \ker(L) = \{x \in \mathcal{M} : T_t(x) = x \text{ for all } t \geq 0\}$$

is a von Neumann subalgebra of $\mathcal{M}$, and each T_t is an $\mathcal{N}$ -bimodule map,

$$T_t(axb) = aT_t(x)b, \quad a, b \in \mathcal{N}, x \in \mathcal{M}.$$

In particular,

$$T_t \circ E = E \circ T_t = E,$$

where $E : \mathcal{M} \rightarrow \mathcal{N}$ is the τ -preserving conditional expectation onto $\mathcal{N}$. Before introducing the Poincaré inequality, we recall the associated energy and gradient forms.

Definition 2.1 (Dirichlet form). *The Dirichlet form (or energy functional) of the semigroup is the quadratic map*

$$\mathcal{E} : \text{Dom}(L^{1/2}) \rightarrow \mathbb{R}_{\geq 0}, \quad \mathcal{E}(x) := \|L^{1/2}x\|_{L^2(\mathcal{M}, \tau)}^2.$$

For $x \in \text{Dom}(L)$, this equals $\tau(x^*Lx)$.

Definition 2.2 (Gradient form). *The noncommutative gradient form associated to the Lindbladian L is the sesquilinear map $\Gamma : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{M}$ where $\mathcal{A} := \text{Dom}(L) \cap \mathcal{M}$,¹ is defined by [15, Section 2.1]*

$$\Gamma(x, y) = \frac{1}{2}(L(x^*)y + x^*L(y) - L(x^*y)), \quad x, y \in \mathcal{A}.$$

Equivalently, by the weak*-limit

$$\Gamma(x, y) := \text{w}^*\text{-}\lim_{t \downarrow 0} \frac{1}{2t} \left(T_t(x^*y) - T_t(x^*)y - x^*T_t(y) + x^*y \right), \quad x, y \in \mathcal{A}, \quad (7)$$

or equivalently from [11, Lemma 9.1],

$$\Gamma(x, y) := \text{w}^*\text{-}\lim_{\epsilon \downarrow 0} \frac{1}{2} \left(x^* \frac{L}{1 + \epsilon L}(y) + \frac{L}{1 + \epsilon L}(x)^*y - \frac{L}{1 + \epsilon L}(x^*y) \right), \quad x, y \in \mathcal{A}. \quad (8)$$

In particular, $\Gamma(x, x) \in \mathcal{M}_+$ for every $x \in \mathcal{A}$. We can also rewrite the energy form as

$$\mathcal{E}(x) = \tau(\Gamma(x, x)), \quad x \in \mathcal{A}.$$

Definition 2.3. *Let $\{T_t\}_{t \geq 0}$ a τ -symmetric QMS with Lindbladian L , gradient form Γ , and $E : \mathcal{M} \rightarrow \mathcal{N}$ be the τ -preserving conditional expectation onto $\mathcal{N}$. Let $p, r/2 \in [1, \infty]$, we say that*

- T_t admits the Poincaré inequality $\text{PI}(p, r)$ with constant $c > 0$, if

$$\|x - E(x)\|_{L^p(\mathcal{M}, \tau)} \leq c \|\Gamma(x, x)\|_{L^{r/2}(\mathcal{M}, \tau)}^{1/2}, \quad x \in \mathcal{A}$$

- L admits the spectral gap $\alpha > 0$ if

$$\alpha \|x - E(x)\|_{L^2(\mathcal{M}, \tau)}^2 \leq \|\Gamma(x, x)\|_{L^1(\mathcal{M}, \tau)}. \quad x \in \mathcal{A}. \quad (9)$$

¹ $\mathcal{A}$ is a dense $*$ -subalgebra of $\mathcal{M}$ and a core for $L^{1/2}$ [11, 12].

Note that the spectral gap is a special case of Poincaré inequalities PI(2, 2), and the first main result we later derive is that given a spectral gap, one can derive a PI(p, p) for $p = 2$ and $p \geq 3$. Integrating Equation (9) gives the L^2 decay estimate

$$\|T_t(x) - E(x)\|_{L^2(\mathcal{M}, \tau)} \leq e^{-\alpha t} \|x - E(x)\|_{L^2(\mathcal{M}, \tau)}, \quad x \in \mathcal{A}, \quad t \geq 0.$$

2.3 Construction of amalgamated free product (AFP)

2.3.1 Motivation: Fock spaces

Starting from a one-particle Hilbert space $\mathcal{H}$ with vacuum vector ξ , one defines three standard Fock space constructions. The symmetric Fock space arises as the GNS representation of the CCR (canonical commutation relations) algebra, the antisymmetric Fock space corresponds to the CAR (canonical anti-commutation relations) algebra, and the full (free) Fock space serves as the analogue without commutation restrictions.

- Bosons (CCR).

$$\mathcal{F}_{\text{Bose}}(\mathcal{H}) = \mathbb{C}\xi \oplus \bigoplus_{n=1}^{\infty} \text{Sym}^n(\mathcal{H}).$$

- Fermions (CAR).

$$\mathcal{F}_{\text{Fermi}}(\mathcal{H}) = \mathbb{C}\xi \oplus \bigoplus_{n=1}^{\infty} \bigwedge^n \mathcal{H}.$$

- Free particles (free product algebra).

$$\mathcal{F}_{\text{Free}}(\mathcal{H}) = \mathbb{C}\xi \oplus \bigoplus_{n=1}^{\infty} \mathcal{H}^{\otimes n}. \quad (10)$$

In the presence of multiple vacua encoded by a subalgebra, one passes to the amalgamated free product (AFP) Fock space, which may be viewed heuristically as obtained by gluing free Fock spaces over the different vacuum sectors.

2.3.2 Amalgamated free products (AFP)

We recall the construction of amalgamated free products (AFP) of tracial von Neumann algebras, following [4, 9, 22, 44, 49]. Let $(\mathcal{M}_1, \tau_1)$ and $(\mathcal{M}_2, \tau_2)$ be tracial von Neumann algebras with a common subalgebra $\mathcal{N} \subset \mathcal{M}_i$, $i = 1, 2$, such that $\tau_1|_{\mathcal{N}} = \tau_2|_{\mathcal{N}}$. There exist unique trace-preserving conditional expectations $E_i: \mathcal{M}_i \rightarrow \mathcal{N}$ [47, Theorem 4.2], and we write

$$\mathcal{M}_i^{\circ} := \ker(E_i).$$

Each $\mathcal{M}_i$ is naturally an $\mathcal{N}$ - $\mathcal{N}$ bimodule with $\mathcal{N}$ -valued inner product $\langle x, y \rangle_{\mathcal{N}} = E_i(x^*y)$ for $x, y \in \mathcal{M}_i$, and hence a Hilbert $\mathcal{N}$ - $\mathcal{N}$ bimodule in the sense of [30]. The orthogonal direct sum $\bigoplus_i \mathcal{M}_i$ is again an $\mathcal{N}$ - $\mathcal{N}$ bimodule with inner product

$\langle \oplus_i x_i, \oplus_i y_i \rangle_{\mathcal{N}} = \sum_i \langle x_i, y_i \rangle_{\mathcal{N}}$. On simple tensors, the interior tensor product $\mathcal{M}_{i_1} \otimes_{\mathcal{N}} \cdots \otimes_{\mathcal{N}} \mathcal{M}_{i_n}$ carries the inner product

$$\langle x_1 \otimes \cdots \otimes x_n, y_1 \otimes \cdots \otimes y_n \rangle_{\mathcal{N}} = E_{i_n} \left(x_n^* E_{i_{n-1}} (\cdots E_{i_1} (x_1^* y_1) \cdots) y_n \right). \quad (11)$$

The AFP $\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$ can be realized as the completion of the $*$ -algebra of reduced words.

$$\mathcal{N} \oplus \bigoplus_{n=1}^{\infty} \bigoplus_{\substack{i_1, \dots, i_n \in \{1, 2\} \\ i_k \neq i_{k+1}}} \mathcal{M}_{i_1} \otimes_{\mathcal{N}} \cdots \otimes_{\mathcal{N}} \mathcal{M}_{i_n}. \quad (12)$$

This decomposition inherits the structure of a Hilbert $\mathcal{N}$ - $\mathcal{N}$ bimodule by discussion around eq. (11).

There are natural embeddings

$$\pi_i : \mathcal{M}_i \hookrightarrow \mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2, \quad i = 1, 2. \quad (13)$$

The canonical trace on the AFP, denoted by $\tilde{\tau} := \tau_1 * \tau_2$, is defined on reduced words. By [47, Theorem 4.2], there exists a unique $\tilde{\tau}$ -preserving conditional expectation $\tilde{E} : \mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2 \rightarrow \mathcal{N}$ such that:

- (i) $\pi_1|_{\mathcal{N}} = \pi_2|_{\mathcal{N}} = \text{Id}_{\mathcal{N}}$;
- (ii) $\tilde{\tau} \circ \tilde{E} = \tilde{\tau}$;
- (iii) $\tilde{E} \circ \pi_i = E_i$, $i = 1, 2$;
- (iv) $\tilde{\tau}|_{\mathcal{N}} = \tau_1|_{\mathcal{N}} = \tau_2|_{\mathcal{N}}$;
- (v) (*Freeness with amalgamation* [49]) if $i_1, \dots, i_n \in \{1, 2\}$ with $i_k \neq i_{k+1}$ and $m_k \in \mathcal{M}_{i_k}$, then

$$\tilde{\tau}(\pi_{i_1}(m_1) \cdots \pi_{i_n}(m_n)) = 0, \quad \tilde{E}(\pi_{i_1}(m_1) \cdots \pi_{i_n}(m_n)) = 0;$$

- (vi) $\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$ is generated by $\pi_1(\mathcal{M}_1) \cup \pi_2(\mathcal{M}_2)$.

The above properties give the abstract characterization of AFP.

The AFP Fock space representation

We also construct the AFP Fock space with non-scalar $\mathcal{N}$. In the scalar case where $\mathcal{N} = \mathbb{C}\xi$, this reduces to the free Fock space representation Equation (10). With the notation above, the GNS Hilbert space of the AFP $(\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2, \tilde{\tau})$ is

$$\mathcal{H}_{12} := \overline{\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2}^{\|\cdot\|_2}, \quad \|x\|_2 = \tilde{\tau}(x^* x)^{1/2}.$$

with scalar inner product induced from the $\mathcal{N}$ -valued inner product:

$$\langle x, y \rangle := \tilde{\tau}(\tilde{E}(x^* y)) = \tilde{\tau}(\langle x, y \rangle_{\mathcal{N}}) \in \mathbb{C}. \quad (14)$$

One can then write $\mathcal{H}_{12}$ ² as [9]:

$$\mathcal{H}_{12} = L^2(\mathcal{N}) \oplus \bigoplus_{n=1}^{\infty} \bigoplus_{\substack{i_1, \dots, i_n \in \{1,2\} \\ i_k \neq i_{k+1}}} L^2(\mathring{\mathcal{M}}_{i_1} \otimes_{\mathcal{N}} \dots \otimes_{\mathcal{N}} \mathring{\mathcal{M}}_{i_n}). \quad (15)$$

where $\oplus$ denotes the orthogonal direct sum with respect to the inner product in (14).

Each inclusion $\pi_i : \mathcal{M}_i \hookrightarrow \mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2 \subset B(\mathcal{H}_{12})$, $i = 1, 2$, defines a left regular action on simple tensors, which form a dense set, given by [9, Lemma 4.7.3]

$$\pi_i(x)(n\xi_{\mathcal{N}}) = (x - E_i(x)) \otimes_{\mathcal{N}} n\xi_{\mathcal{N}} + E_i(x)n\xi_{\mathcal{N}}, \quad n \in \mathcal{N},$$

where $\xi_{\mathcal{N}}$ denotes the GNS vector for $L^2(\mathcal{N}) = \overline{\mathcal{N}\xi_{\mathcal{N}}}$. For $\zeta_1 \otimes \dots \otimes \zeta_n \in \mathring{\mathcal{M}}_{i_1} \otimes_{\mathcal{N}} \dots \otimes_{\mathcal{N}} \mathring{\mathcal{M}}_{i_n}$, $i_1, \dots, i_n \in \{1, 2\}$,

$$\pi_i(x)(\zeta_1 \otimes \dots \otimes \zeta_n) = \begin{cases} (x\zeta_1 - E_i(x\zeta_1)) \otimes \zeta_2 \otimes \dots \otimes \zeta_n + E_i(x\zeta_1)\zeta_2 \otimes \dots \otimes \zeta_n, & i_1 = i, \\ (x - E_i(x)) \otimes \zeta_1 \otimes \dots \otimes \zeta_n + E_i(x)\zeta_1 \otimes \dots \otimes \zeta_n, & i_1 \neq i, \end{cases} \quad (16)$$

In words, if the leading index equals i , $\pi_i(x)$ acts by left multiplication on the first tensor leg, with the usual centering correction on $\mathring{\mathcal{M}}_i$; otherwise, it creates a new tensor leg on the left up to centering correction³. Each π_i is a faithful normal $*$ -homomorphism with $\pi_i|_{\mathcal{N}} = \text{Id}_{\mathcal{N}}$. Thus eq. (13) extends to the *GNS representation* after completion.

3 Poincaré inequality for trace-symmetric QMS

In this section we prove Theorem 1.2. The argument proceeds as follows: We introduce the swap-operator construction as a noncommutative symmetrization, state auxiliary lemmas for the associated derivation, establish a generalized Klein inequality giving a convex chain rule, and finally prove the $\text{PI}(p, p)$ inequality and its non-self-adjoint extension. To ease notation, in this section, we write p -norms $\|\cdot\|_{L^p(\mathcal{M}, \tau)}$ as $\|\cdot\|_p$.

3.1 Swap operator construction

Specifically, we are working in the case where $(\mathcal{M}_1, \tau_1)$ and $(\mathcal{M}_2, \tau_2)$ are two copies of $(\mathcal{M}, \tau)$ containing the von Neumann subalgebra $\mathcal{N}$. We use the shorthand $\mathcal{H}_{ij} := L^2(\mathcal{M}_i *_{\mathcal{N}} \mathcal{M}_j, \tilde{\tau})$ and define a swap operator

$$U : \mathcal{H}_{12} \rightarrow \mathcal{H}_{21}$$

²This construction is equivalent to introducing Connes fusion [47], but we avoid introducing this machinery here for simplicity.

³Under the identification obtained by regrouping the decomposition (12) of $\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$ as $\mathcal{M}_i \otimes_{\mathcal{N}} \dots$, the representations coincide with the left regular actions: $\pi_i(x) = x \otimes \mathbf{1}$.

that exchanges the roles of the two legs $1 \leftrightarrow 2$ in the Hilbert space generated by the AFP $\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$. As a prelude, we first display the commuting diagram that organizes the construction. The swap unitary U implementing Ad_U is constructed later in this subsection.

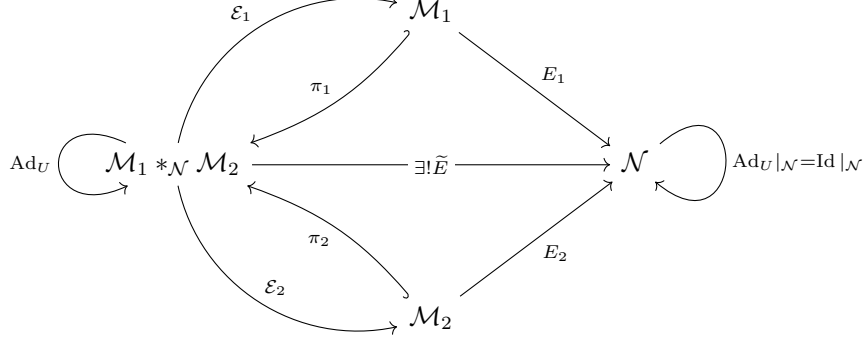


Fig. 2 Roadmap for the AFP construction and the swap automorphism.

Here, each arrow denotes

- (i) $\pi_i: \mathcal{M}_i \hookrightarrow \mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$ is the canonical inclusion.
- (ii) $\mathcal{E}_i: \mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2 \rightarrow \mathcal{M}_i$ is the canonical conditional expectation onto $\mathcal{M}_i$, characterized by

$$\mathcal{E}_i \circ \pi_i = \text{Id}_{\mathcal{M}_i}, \quad \mathcal{E}_i(\pi_j(x)) = E_j(x) \quad (j \neq i).$$

- (iii) $E_i: \mathcal{M}_i \rightarrow \mathcal{N}$ is the given τ_i -preserving conditional expectation.
- (iv) The two triangles commute, i.e. $E_i \circ \mathcal{E}_i = \tilde{E}$.
- (v) $\tilde{E}: \mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2 \rightarrow \mathcal{N}$ is the unique conditional expectation whose restriction to $\pi_i(\mathcal{M})$ equals E_i .
- (vi) $\text{Ad}_U: \mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2 \rightarrow \mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$ is the spatial $*$ -automorphism implemented by the swap unitary U , satisfying

$$\text{Ad}_U \circ \pi_1 = \pi_2, \quad \text{Ad}_U \circ \pi_2 = \pi_1, \quad \text{Ad}_U|_{\mathcal{N}} = \text{Id}|_{\mathcal{N}}.$$

Expanding the canonical orthogonal decomposition eq. (15),

$$\begin{aligned} \mathcal{H}_{12} = & L^2(\mathcal{N}) \oplus (L^2(\mathring{\mathcal{M}}_1) \oplus L^2(\mathring{\mathcal{M}}_2)) \\ & \oplus (L^2(\mathring{\mathcal{M}}_1 \otimes_{\mathcal{N}} \mathring{\mathcal{M}}_2) \oplus L^2(\mathring{\mathcal{M}}_2 \otimes_{\mathcal{N}} \mathring{\mathcal{M}}_1)) \\ & \oplus (L^2(\mathring{\mathcal{M}}_1 \otimes_{\mathcal{N}} \mathring{\mathcal{M}}_2 \otimes_{\mathcal{N}} \mathring{\mathcal{M}}_1) \oplus \cdots). \end{aligned}$$

Similarly, the decomposition of $\mathcal{H}_{21}$ swaps $1 \leftrightarrow 2$. Both decompositions are canonically isometric. In the symmetric case $\mathcal{M}_1 = \mathcal{M}_2 = \mathcal{M}$, the swap operator is defined by

$$\begin{aligned} U: L^2(\mathring{\mathcal{M}}_{i_1} \otimes_{\mathcal{N}} \cdots \otimes_{\mathcal{N}} \mathring{\mathcal{M}}_{i_n}) &\longrightarrow L^2(\mathring{\mathcal{M}}_{3-i_1} \otimes_{\mathcal{N}} \cdots \otimes_{\mathcal{N}} \mathring{\mathcal{M}}_{3-i_n}), \\ \eta_1 \otimes \cdots \otimes \eta_n &\longmapsto \zeta_1 \otimes \cdots \otimes \zeta_n, \end{aligned}$$

with $\zeta_k = \eta_k$ regarded in the corresponding copy of $\mathring{\mathcal{M}}_{3-i_k}$. On the vacuum sector we set $U|_{L^2(\mathcal{N})} = \text{Id}|_{L^2(\mathcal{N})}$. This implies a unitary isomorphism $\mathcal{H}_{12} \simeq \mathcal{H}_{21}$ implementing the label exchange $1 \leftrightarrow 2$.

Proposition 3.1. *The swap operator U is unitary.*

Proof. It suffices to verify norm preservation on simple tensors. Let

$$\eta = \eta_1 \otimes \cdots \otimes \eta_n \in \mathring{\mathcal{M}}_{i_1} \otimes_{\mathcal{N}} \cdots \otimes_{\mathcal{N}} \mathring{\mathcal{M}}_{i_n},$$

and set $\zeta_k = \eta_k \in \mathring{\mathcal{M}}_{3-i_k}$. Using the L^2 inner product eq. (14) and $\tilde{\tau}|_{\mathcal{N}} = \tau|_{\mathcal{N}}$,

$$\begin{aligned} \|U\eta\|_2^2 &= \langle U\eta, U\eta \rangle = \tilde{\tau}(\tilde{E}((U\eta)^*(U\eta))) \\ &= \tau(E_{3-i_n}(\zeta_n^* E_{3-i_{n-1}}(\cdots E_{3-i_1}(\zeta_1^* \zeta_1) \cdots) \zeta_n)) \\ &= \tau(E_{i_n}(\eta_n^* E_{i_{n-1}}(\cdots) \eta_n)) = \|\eta\|_2^2, \end{aligned}$$

where the last equality follows by relabeling $3 - i_k \mapsto i_k$. Thus U extends to an isometry $\mathcal{H}_{12} \rightarrow \mathcal{H}_{21}$. The inverse construction gives an isometry $\mathcal{H}_{21} \rightarrow \mathcal{H}_{12}$, hence U is unitary. $\square$

Proposition 3.2. *Let U be the swap unitary on $\mathcal{H}_{12}$. Then $\text{Ad}_U(T) = UTU^*$ restricts to a $*$ -automorphism of $\mathcal{M}_1 *_N \mathcal{M}_2$ satisfying:*

- (i) $\text{Ad}_U(\pi_i(x)) = \pi_{3-i}(x)$ for all $x \in \mathcal{M}$.
- (ii) Ad_U commutes with $\tilde{E}$.
- (iii) Ad_U preserves $\tilde{\tau}$.

Proof. Since U is unitary, Ad_U defines a spatial $*$ -isomorphism between $B(\mathcal{H}_{12})$ and $B(\mathcal{H}_{21})$ [46, Section 3]. Identifying $\mathcal{H}_{21}$ with $\mathcal{H}_{12}$ via U , we view Ad_U as an automorphism of $B(\mathcal{H}_{12})$.

(i) It suffices to check the action on simple tensors. Let $\eta = \eta_1 \otimes \cdots \otimes \eta_n$ with $i_1 = i$. Using eq. (16) and $U|_{L^2(\mathcal{N})} = \text{Id}$,

$$\begin{aligned} U\pi_i(x)\eta &= (x\zeta_1 - E_{3-i}(x\zeta_1)) \otimes \cdots \otimes \zeta_n + E_{3-i}(x\zeta_1)\zeta_2 \otimes \cdots \otimes \zeta_n \\ &= \pi_{3-i}(x)(\zeta_1 \otimes \cdots \otimes \zeta_n) = \pi_{3-i}(x)U\eta, \end{aligned}$$

where $\zeta_k = \eta_k$ viewed in $\mathring{\mathcal{M}}_{3-i_k}$. Thus $\text{Ad}_U(\pi_i(x)) = \pi_{3-i}(x)$. Since $\mathcal{M}_1 *_N \mathcal{M}_2$ is generated by $\pi_1(\mathcal{M})$ and $\pi_2(\mathcal{M})$, Ad_U restricts to a $*$ -automorphism.

(ii) Following Figure 2,

$$\mathrm{Ad}_U \circ \tilde{E} \circ \pi_i = \tilde{E} \circ \pi_i = \tilde{E} \circ \pi_{3-i} = \tilde{E} \circ \mathrm{Ad}_U \circ \pi_i,$$

since $U|_{L^2(\mathcal{N})} = \mathrm{Id}|_{L^2(\mathcal{N})}$ and $\mathcal{M}_1 = \mathcal{M}_2 = \mathcal{M}$. As $\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$ is generated by $\pi_1(\mathcal{M})$ and $\pi_2(\mathcal{M})$ and both Ad_U and $\tilde{E}$ are normal, the relation extends to all of $\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$. Thus $\mathrm{Ad}_U \circ \tilde{E} = \tilde{E} = \tilde{E} \circ \mathrm{Ad}_U$. (iii) Because $\tilde{\tau} = \tilde{\tau} \circ \tilde{E}$,

$$\tilde{\tau}(\mathrm{Ad}_U(T)) = \tilde{\tau}(\tilde{E}(\mathrm{Ad}_U(T))) = \tilde{\tau}(\tilde{E}(T)) = \tilde{\tau}(T), \quad T \in \mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2.$$

Hence Ad_U swaps the two embeddings, commutes with $\tilde{E}$, and preserves $\tilde{\tau}$. $\square$

We will see in Section 3.4 that the swap unitary U is the essential ingredient for the symmetrization technique, replacing the product measure in the commutative case [21, Section 4.1].

3.2 Auxiliary lemmas

We first show several auxiliary lemmas used in the proof of Theorem 1.2. They state the relevant estimates in terms of a noncommutative derivation, replacing the product-measure construction of doubling-up variables techniques from [21] in the semi-commutative case. We define the noncommutative derivation

$$\delta(x) := \pi_1(x) - \pi_2(x) \in \mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2.$$

Lemma 3.3. *Let $x \in \mathcal{M}$. Then for any $1 \leq p < \infty$,*

$$\|x - E_1(x)\|_p^p \leq \|\delta(x)\|_p^p.$$

Proof. Since the conditional expectation $\mathcal{E}_1 : L_p(\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2) \rightarrow L_p(\mathcal{M}_1)$ is contractive [47], we have

$$\|\mathcal{E}_1(\delta(x) - E(\delta(x)))\|_p \leq \|\delta(x) - E(\delta(x))\|_p.$$

Moreover,

$$\mathcal{E}_1(\delta(x)) = \mathcal{E}_1(\pi_1(x)) - \mathcal{E}_1(\pi_2(x)) = x - E_1(x),$$

and since $E \circ \pi_i = E_i$ with $E_1 = E_2$, one has $E(\delta(x)) = 0$. The claim follows. $\square$

The following lemma gives the noncommutative version of the symmetrization technique:

Lemma 3.4. *Let $x \in \mathcal{M}_{\mathrm{sa}}$, and let $\delta(x)$ be as above and U the swap unitary satisfying $\mathrm{Ad}_U \circ \pi_1 = \pi_2$ and $\mathrm{Ad}_U \circ \pi_2 = \pi_1$, let $h : \mathbb{R} \rightarrow \mathbb{R}$ be an odd function. Then*

- (i) $\mathrm{Ad}_U(h(\delta(x))) = -h(\delta(x))$,
- (ii) $\tilde{\tau}(h(\delta(x))) = 0$,
- (iii) $E(h(\delta(x))) = 0$,

where $\tilde{\tau}$ is the canonical trace on $\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$.

Proof. By Theorem 3.2, $\text{Ad}_U(\delta(x)) = -\delta(x)$. Since h is odd, $h(-\delta(x)) = -h(\delta(x))$. The U -invariance of $\tilde{\tau}$ and E then implies the stated identities. $\square$

Lemma 3.5. *Let $\tilde{L}$ and $\tilde{\Gamma}$ be the Lindblad generator and gradient form on $\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$ induced from $\mathcal{M}$ as in [8], characterized by*

$$\tilde{L}(\pi_i(x)) = \pi_i(L_i(x)), \quad \tilde{\Gamma}(\pi_i(x), \pi_i(x)) = \pi_i(\Gamma_i(x, x)), \quad i = 1, 2,$$

where $L_i = L$ and $\Gamma_i = \Gamma$ are the corresponding operators on $\mathcal{M}$. Then for every $x \in \mathcal{M}$ and $p \geq 2$,

$$\|\tilde{\Gamma}(\delta(x), \delta(x))\|_{p/2} \leq 2\|\Gamma(x, x)\|_{p/2}.$$

Proof. A direct computation shows

$$\tilde{\Gamma}(\delta(x), \delta(x)) = \pi_1(\Gamma(x, x)) + \pi_2(\Gamma(x, x)).$$

Since π_i preserves the $L^{p/2}$ -norm and the trace, the inequality follows immediately. $\square$

3.3 Generalized Klein inequalities

The key analytic input for the tracial PI(p, p) inequality is a convexity estimate relating functional calculus and the gradient form. This is provided by Markov dilation (Appendix A) and a generalized Klein inequality, which is used to obtain a convex-chain rule for the Dirichlet form. We recall the tracial version below and derive the corresponding gradient estimate.

Lemma 3.6 (Generalized Klein inequality for tracial von Neumann algebra). *Let $(\mathcal{M}, \tau)$ be a von Neumann algebra as in the last section. Let $x, y \in \mathcal{M}_{\text{sa}}$ be bounded self-adjoint elements, and let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function whose squared derivative $\psi = (\varphi')^2$ is convex on $\mathbb{R}$. Then*

$$\tau[(\varphi(x) - \varphi(y))^2] \leq \frac{1}{2}\tau[(x - y)^2(\psi(x) + \psi(y))].$$

Proof. By [39, Prop. 3], the claim holds in finite dimensions, and the general bounded case follows by spectral approximation and norm continuity. $\square$

Proposition 3.7 (Convex-chain estimate for gradient form). *Assume the hypotheses of Section 2 and lemma 3.6, for every self-adjoint element $x \in \text{Dom}(L) \subset L_2(\mathcal{M}, \tau)$,*

$$\mathcal{E}(\varphi(x)) \leq \tau(\Gamma(x, x)\psi(x)),$$

where $\mathcal{E}(f) := \tau(\Gamma(f, f))$ is the Dirichlet form associated with L .

We defer the proof to Appendix A, where the main technique is Markov dilation that generalizes the Markov processes in [21]. We use the main dilation theorem of [27], which states that every weak*-continuous semigroup of normal, unital, completely positive maps that is self-adjoint on $L^2(\mathcal{M}, \tau)$ admits a Markov dilation in the sense of Definition A.1.

3.4 Proof of trace-symmetric PI(p, p)

We now prove Theorem 1.2. We begin by recalling its statement.

Theorem. *Let $(\mathcal{M}, \tau)$ be a tracial von Neumann algebra and $(T_t)_{t \geq 0} = e^{-tL}$ a τ -symmetric quantum Markov semigroup with fixed-point algebra $\mathcal{N}$ and conditional expectation $E : \mathcal{M} \rightarrow \mathcal{N}$. If $\{T_t\}_{t \geq 0}$ has a spectral gap $\alpha > 0$, then for every self-adjoint x in the domain of L and $p = 2$ or $p \geq 3$, it satisfies the Poincaré inequality PI(p, p) with constant $\frac{p}{\sqrt{2\alpha}}$, i.e.,*

$$\|x - E(x)\|_p \leq \frac{p}{\sqrt{2\alpha}} \|\Gamma(x, x)^{1/2}\|_p. \quad (\text{PI}(p, p))$$

Proof. For $x \in \mathcal{M}_{sa}$, Lemma 3.3 gives

$$\|x - E_1(x)\|_p^p \leq \|\delta(x)\|_p^p, \quad \delta(x) := \pi_1(x) - \pi_2(x) \in (\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2)_{sa}. \quad (17)$$

Define the odd function $\varphi(s) := \text{sgn}(s)|s|^{p/2}$ (cf. [21]). Its squared derivative is

$$\psi(s) := (\varphi'(s))^2 = \left(\frac{p}{2}\right)^2 |s|^{p-2},$$

which is convex for $p = 2$ and $p \geq 3$. Since φ is odd, Lemma 3.4 implies $E(\varphi(\delta(x))) = 0$, and hence

$$\|\varphi(\delta(x)) - E(\varphi(\delta(x)))\|_2^2 = \tilde{\tau}(\varphi(\delta(x))^* \varphi(\delta(x))) = \|\delta(x)\|_p^p.$$

With the spectral-gap assumption (9) (extended to $\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$ by Theorem B.3),

$$\alpha \|\delta(x)\|_p^p \leq \|\tilde{\Gamma}(\varphi(\delta(x)), \varphi(\delta(x)))^{1/2}\|_2^2 = \tilde{\tau}(\varphi(\delta(x))^* \tilde{L}(\varphi(\delta(x)))) =: \mathcal{E}(\varphi(\delta(x))).$$

By Theorem 3.7 (convex chain rule),

$$\mathcal{E}(\varphi(\delta(x))) \leq \tilde{\tau}(\tilde{\Gamma}(\delta(x), \delta(x))\psi(\delta(x))) = \left(\frac{p}{2}\right)^2 \tilde{\tau}(\tilde{\Gamma}(\delta(x), \delta(x))|\delta(x)|^{p-2}).$$

Applying Hölder's inequality,

$$\alpha \|\delta(x)\|_p^p \leq \left(\frac{p}{2}\right)^2 \tilde{\tau}(\tilde{\Gamma}(\delta(x), \delta(x))^{p/2})^{2/p} \|\delta(x)\|_p^{p-2}.$$

Dividing by $\|\delta(x)\|_p^{p-2}$ and applying Lemma 3.5 gives

$$\|\delta(x)\|_p^2 \leq \frac{p^2}{4\alpha} \tilde{\tau}(\tilde{\Gamma}(\delta(x), \delta(x))^{p/2})^{2/p} \leq \frac{p^2}{2\alpha} \tau(\Gamma(x, x)^{p/2})^{2/p}.$$

Taking square roots and using (17) completes the proof:

$$\|x - E(x)\|_p \leq \frac{p}{\sqrt{2\alpha}} \|\Gamma(x, x)^{1/2}\|_p.$$

□

Remark 3.8 (Missing Parameters [21]). *Theorem 1.2 also holds for $p \in (2, 3)$ with an extra factor of $\sqrt{2}$ on the right hand side through applying a variant of Lemma 3.6 that only needs ψ to be monotone.*

We can also generalize to non-self-adjoint elements in $\mathcal{M}$ by the following corollary. We note that in the non-tracial setting, the one-sided estimate in Equation (25) may be useful in the analysis of quantum algorithms.

Corollary 3.9 (Non-self-adjoint PI(p, p)). *Under the assumptions of Theorem 1.2, for every $p = 2$ and $p \geq 3$ and $x \in \mathcal{M}$:*

$$\|x - E(x)\|_p \leq \frac{p}{\sqrt{2\alpha}} (\|\Gamma(x, x)\|_{p/2}^{1/2} + \|\Gamma(x^*, x^*)\|_{p/2}^{1/2}).$$

Proof. Since $x = \operatorname{Re} x + i \operatorname{Im} x$ with $\operatorname{Re} x, \operatorname{Im} x \in \mathcal{M}_{sa}$, the triangle inequality in L_p gives

$$\|x - E(x)\|_p \leq \|\operatorname{Re} x - E(\operatorname{Re} x)\|_p + \|\operatorname{Im} x - E(\operatorname{Im} x)\|_p.$$

We apply Theorem 1.2 to $\operatorname{Re} x$ and $\operatorname{Im} x$,

$$\|\operatorname{Re} x - E(\operatorname{Re} x)\|_p + \|\operatorname{Im} x - E(\operatorname{Im} x)\|_p \leq \frac{p}{\sqrt{2\alpha}} (\|\Gamma(\operatorname{Re} x, \operatorname{Re} x)\|_{p/2}^{1/2} + \|\Gamma(\operatorname{Im} x, \operatorname{Im} x)\|_{p/2}^{1/2}).$$

Since

$$\|u\|_{\Gamma, p} := \|\Gamma(u, u)\|_{p/2}^{1/2}$$

is a seminorm and

$$\operatorname{Re} x = \frac{x + x^*}{2}, \quad \operatorname{Im} x = \frac{x - x^*}{2i},$$

its triangle inequality gives

$$\|\Gamma(\operatorname{Re} x, \operatorname{Re} x)\|_{p/2}^{1/2} + \|\Gamma(\operatorname{Im} x, \operatorname{Im} x)\|_{p/2}^{1/2} \leq \|\Gamma(x, x)\|_{p/2}^{1/2} + \|\Gamma(x^*, x^*)\|_{p/2}^{1/2}.$$

Combining the above estimates gives the stated corollary. □

4 Poincaré inequality for GNS-detailed balance QMS

Previously, we studied trace-symmetric quantum Markov semigroups and proved the $\text{PI}(p, p)$ inequality (theorem 1.2) under the assumptions of a spectral gap on a tracial von Neumann algebra $(\mathcal{M}, \tau)$. In this section, we extend this result to von Neumann algebras that are not equipped with a trace, and to QMSs satisfying the corresponding GNS detailed-balance condition with respect to a non-tracial state. The main tools are Haagerup's reduction method [19] and Kosaki's L^p interpolation theorem [28]. We also formulate the results using Haagerup's functorial L^p construction. To set notation, we first recall results from Haagerup's L^p spaces and Kosaki's L^p interpolation spaces.

4.1 Haagerup's and Kosaki's L^p spaces

4.1.1 Haagerup's L^p spaces

We give a concise introduction to Haagerup's L^p spaces [19, Section 1.2][18] for general von Neumann algebras. Let $\mathcal{M}$ be a von Neumann algebra and ϕ a faithful normal semifinite weight on $\mathcal{M}$. Denote by $\sigma_t^\phi = \sigma_t$ ($t \in \mathbb{R}$) its modular automorphism group, and form the crossed product [47, Section X], $\mathcal{R} = \mathcal{M} \rtimes_\sigma \mathbb{R} = \{\pi_\sigma(\mathcal{M}) \cup \lambda(\mathbb{R})\}''$, where π_σ and λ act on $L^2(\mathcal{M}, \phi) \otimes L^2(\mathbb{R}, dt)$. It is well known that $\mathcal{R}$ is semifinite and admits a unique normal semifinite faithful trace τ (the *canonical trace*) characterized by the scaling

$$\tau \circ \widehat{\sigma}_t = e^{-t} \tau, \quad t \in \mathbb{R}, \quad (18)$$

where $\widehat{\sigma}$ is the dual action on $\mathcal{R}$.

We define and summarize some useful properties of Haagerup's L^p -spaces [18].

Definition 4.1 (Haagerup's L^p Spaces). *With the notation above, for $0 < p \leq \infty$, let $\widetilde{\mathcal{R}}$ be the space of τ -measurable⁴ operators affiliated with $\mathcal{R}$. For $0 < p \leq \infty$, the Haagerup L^p -space is defined by*

$$L^p(\mathcal{M}) := \{x \in \widetilde{\mathcal{R}} \mid \widehat{\sigma}_t(x) = e^{-t/p} x, \quad t \in \mathbb{R}\}.$$

Proposition 4.2. *From Theorem 4.1 of the Haagerup $L^p(\mathcal{M})$, we have the following properties:*

- (i) $L^p(\mathcal{M})$ is a closed, self-adjoint $\mathcal{M}$ -bimodule in $\widetilde{\mathcal{R}}$, stable under $x \mapsto x^*$ and $x \mapsto |x|$.
- (ii) $L^p(\mathcal{M})$ is linearly spanned by the positive part $L^p(\mathcal{M})_+ := L^p(\mathcal{M}) \cap \widetilde{\mathcal{R}}_+$.
- (iii) $L^\infty(\mathcal{M}) = \mathcal{M}$, since the fixed-point algebra of $\widehat{\sigma}$, $\mathcal{R}^{\widehat{\sigma}} = \pi_\sigma(\mathcal{M}) \cong \mathcal{M}$.
- (iv) The map $\mathcal{M}_*^+ \ni \omega \mapsto h_\omega \in L^1(\mathcal{M})_+$ is an isometric isomorphism, and the dual weight $\widehat{\omega}$ on $\mathcal{R}$ is equal to $\tau(h_\omega \cdot)$.

⁴An operator a on $L^2(\mathcal{M}, \phi) \otimes L^2(\mathbb{R}, dt)$ affiliated with $\mathcal{R}$ is called τ -measurable [48] if it is measurable with respect to $(\mathcal{R}, \tau)$.

(v) By linear extension of the above bijection, we have an isometric isomorphism $\text{tr} : \mathcal{M}_* \ni \omega \mapsto h_\omega \in L^1(\mathcal{M})$, by $\text{tr}(h_\omega) := \omega(1)$. Then we have

$$\|h_\omega\|_1 := \text{tr}(|h_\omega|) = \text{tr}(h_{|\omega|}) = |\omega|(1) = \|\omega\|.$$

(vi) Let $x \in \widetilde{\mathcal{R}}$ with polar decomposition $x = u|x|$, for every $p \in [1, \infty)$, $x \in L^p(\mathcal{M})$ if and only if $|x|^p \in L^1(\mathcal{M})$ and $u \in \mathcal{M}$. Define

$$\|x\|_{L^p(\mathcal{M})} := \left\| |x|^p \right\|_1^{1/p} \quad (0 < p < \infty), \quad \|x\|_\infty := \|x\|.$$

(vii) The multiplication in $\widetilde{\mathcal{R}}$, restricts to a bounded bilinear map

$$L^p(\mathcal{M}) \times L^s(\mathcal{M}) \longrightarrow L^r(\mathcal{M}), \quad \frac{1}{p} + \frac{1}{s} = \frac{1}{r},$$

is of norm 1. Equivalently, Hölder's inequality extends to Haagerup's L^p spaces. In particular, if $1/p + 1/q = 1$, then the pairing

$$L^p(\mathcal{M}) \times L^q(\mathcal{M}) \ni (x, y) \mapsto \text{tr}(xy) \in \mathbb{C}$$

identifies $L^q(\mathcal{M})$ isometrically with the dual of $L^p(\mathcal{M})$ (if $p \neq \infty$), and one has the tracial property

$$\text{tr}(xy) = \text{tr}(yx), \quad x \in L^p(\mathcal{M}), y \in L^q(\mathcal{M}).$$

Remark 4.3. $L^p(\mathcal{M})$ is independent of the choice of the weight ϕ [18, 48] (up to canonical isometric isomorphism), so it is customary to write simply $L^p(\mathcal{M})$. When $\phi = \tau_0$ is a trace, one recovers the Dixmier–Segal noncommutative L^p spaces $L^p(\mathcal{M}, \tau_0)$.

4.1.2 Kosaki's L^p spaces

In this section, we give a brief introduction to Kosaki's L^p spaces based on [20, Appendix A.8], where the noncommutative L^p spaces are defined as complex interpolation spaces between $\mathcal{M}$ and $L^1(\mathcal{M})$. It also turns out that there is a natural isomorphism [28, Theorem 9.1] of Kosaki's and Haagerup's L^p spaces. Using the same notation as the previous paragraph with an extra constraint of $\mathcal{M}$ σ -finite and ϕ be a faithful state so that $\phi(\cdot) = \text{tr}(h_\phi \cdot)$, the first step towards interpolation is to construct a pair of two compatible Banach spaces from $\mathcal{M}$ and $\mathcal{M}_*$. One choice is the injective maps

$$\iota_\eta^\phi : \mathcal{M} \ni a \mapsto h_\phi^\eta a h_\phi^{1-\eta} \in L^1(\mathcal{M}) \quad \text{for } 0 \leq \eta \leq 1.$$

We drop the subscript ϕ whenever the state is clear from context. Define the norm $\|\iota_\eta(a)\| := \|a\|$ on $\iota_\eta(\mathcal{M}) \subset L^1(\mathcal{M})$, so that $\mathcal{M} \cong \iota_\eta(\mathcal{M})$. Therefore, the pair $(\iota_\eta(\mathcal{M}), L^1(\mathcal{M}))$ form a pair of compatible Banach spaces.

Definition 4.4 (Kosaki's L^p spaces [20, Def. A.60]). *With the constructions as above, let $1 < p < \infty$ and $0 \leq \eta \leq 1$. Kosaki's L^p space $L^p(\mathcal{M}, \phi)_\eta$ with respect to ϕ is defined to be the complex interpolation space [28, Section I.1]*

$$C_{1/p}(\iota_\eta(\mathcal{M}), L^1(\mathcal{M})),$$

equipped with the complex interpolation norm $\|\cdot\|_{p,\phi,\eta} (= \|\cdot\|_{C_{1/p}})$. Moreover, one sets

$$L^1(\mathcal{M}, \phi)_\eta := L^1(\mathcal{M}), \quad L^\infty(\mathcal{M}, \phi)_\eta := \iota_\eta(\mathcal{M}).$$

In particular, $L^p(\mathcal{M}, \phi)_0, L^p(\mathcal{M}, \phi)_1, L^p(\mathcal{M}, \phi)_{1/2}$ are called the *left*, *right* and *symmetric* L^p -spaces, respectively. We also establish the isomorphism of different L^p spaces.

Theorem 4.5 (Isomorphism of Kosaki's and Haagerup's L^p spaces). *Let $1 \leq p \leq \infty$ and let q be its conjugate exponent, $1/p + 1/q = 1$. Then for each $0 \leq \eta \leq 1$ one has the identification*

$$L^p(\mathcal{M}, \phi)_\eta = h_\phi^{\eta/q} L^p(\mathcal{M}) h_\phi^{(1-\eta)/q} \subset L^1(\mathcal{M}),$$

and the map

$$\iota_{\eta,p} : L^p(\mathcal{M}) \longrightarrow L^p(\mathcal{M}, \phi)_\eta, \quad a \mapsto h_\phi^{\eta/q} a h_\phi^{(1-\eta)/q}, \quad (19)$$

is an isometry. Equivalently, for all $a \in L^p(\mathcal{M})$,

$$\|h_\phi^{\eta/q} a h_\phi^{(1-\eta)/q}\|_{p,\phi,\eta} = \|a\|_{L^p(\mathcal{M})}.$$

In particular, $L^p(\mathcal{M}) \cong L^p(\mathcal{M}, \phi)_\eta$ via this isometry. On the other hand, $a = \iota_{\eta,q}(x) = h_\phi^{\eta/p} x h_\phi^{(1-\eta)/p}$ is dense in $L_p(\mathcal{M})$ for $x \in \mathcal{M}$, so we have

$$\|\iota_{\eta,p}(a)\|_{p,\phi,\eta} = \|\iota_\eta(x)\|_{p,\phi,\eta} = \|a\|_{L^p(\mathcal{M})} = \|h_\phi^{\eta/p} x h_\phi^{(1-\eta)/p}\|_{L^p(\mathcal{M})}.$$

In particular, when $\eta = 0, 1, \frac{1}{2}$ we recover Kosaki's left, right, and symmetric spaces:

$$\begin{aligned} L^p(\mathcal{M}, \phi)_L &:= L^p(\mathcal{M}) h_\phi^{1/q}, & L^p(\mathcal{M}, \phi)_R &:= h_\phi^{1/q} L^p(\mathcal{M}), \\ L^p(\mathcal{M}, \phi) &:= L^p(\mathcal{M}, \phi)_{1/2} = h_\phi^{1/2q} L^p(\mathcal{M}) h_\phi^{1/2q}, \end{aligned} \quad (20)$$

with norms

$$\|a h_\phi^{1/q}\|_{p,\phi,0} = \|h_\phi^{1/q} a\|_{p,\phi,1} = \|h_\phi^{1/2q} a h_\phi^{1/2q}\|_{p,\phi,1/2} = \|a\|_{L^p(\mathcal{M})}, \quad a \in L^p(\mathcal{M}).$$

With the above identifications in mind, we also summarize the norms in Table 1.

Spaces	Norms
Haagerup's $L^p(\mathcal{M})$	$\ a\ _{L^p(\mathcal{M})} = \ a ^p \ _1^{1/p}$
Kosaki's space $L^p(\mathcal{M}, \phi)_\eta$	$\ \iota_\eta(x)\ _{p, \phi, \eta} = \ h_\phi^{\eta/p} x h_\phi^{(1-\eta)/p}\ _{L^p(\mathcal{M})}$
Left $L^p(\mathcal{M}, \phi)_L$	$\ x h_\phi\ _{p, \phi, 0} = \ x h_\phi^{1/p}\ _{L^p(\mathcal{M})}$
Right $L^p(\mathcal{M}, \phi)_R$	$\ h_\phi x\ _{p, \phi, 1} = \ h_\phi^{1/p} x\ _{L^p(\mathcal{M})}$
Symmetric $L^p(\mathcal{M}, \phi)_{1/2}$	$\ h_\phi^{1/2} x h_\phi^{1/2}\ _{p, \phi, 1/2} = \ h_\phi^{\frac{1}{2p}} x h_\phi^{\frac{1}{2p}}\ _{L^p(\mathcal{M})}$

Table 1 Norms in Haagerup's and Kosaki's L^p spaces for $a \in L^p(\mathcal{M})$ and $x \in \mathcal{M}$.

4.2 GNS and KMS detailed balance with respect to a state

Throughout this section, we work in the GNS representation of $\mathcal{M}$ associated with a normal, faithful state ϕ , and use the notation for the modular operator Δ , modular conjugation J , and modular automorphism group $(\sigma_t^\phi)_{t \in \mathbb{R}}$ introduced in Section 2.1. There are two symmetry conditions [16, Definition 4.1] for QMS relative to the state ϕ .

Definition 4.6 (GNS and KMS detailed balance). *A quantum Markov semigroup $(T_t)_{t \geq 0} : \mathcal{M} \rightarrow \mathcal{M}$ is Gelfand-Naimark-Segal detailed balance with respect to ϕ (GNS- ϕ -detailed balance) if*

$$\phi(T_t(x)^*y) = \phi(x^*T_t(y)), \quad x, y \in \mathcal{M}, t \geq 0;$$

We say T_t is Kubo-Martin-Schwinger detailed balance with respect to ϕ (KMS- ϕ -detailed balance) if

$$\langle \Delta^{\frac{1}{4}} x \xi_\phi, \Delta^{\frac{1}{4}} T_t(y) \xi_\phi \rangle = \langle \Delta^{\frac{1}{4}} T_t(x) \xi_\phi, \Delta^{\frac{1}{4}} y \xi_\phi \rangle, \quad x, y \in \mathcal{M}, t \geq 0. \quad (21)$$

Correspondingly, we call the pre-adjoint $(T_t)_ : \mathcal{M}_* \rightarrow \mathcal{M}_*$ a GNS- or KMS- ϕ -detailed balance quantum channel.*

To fix ideas, we restrict attention to the case where $\mathcal{M}$ has a normal, faithful, finite trace τ . The general von Neumann-algebra case follows after the necessary adaptations of notation (see Section 4.1.1). In the tracial case, every normal state ϕ admits a density $D_\phi \in L^1(\mathcal{M})$ such that $\phi(x) = \tau(D_\phi x)$ for $x \in \mathcal{M}$.

With this choice, setting $x \in \mathcal{M}$, the modular operator can be written as

$$\Delta(x) = D_\phi x D_\phi^{-1},$$

and for $s \in \mathbb{R}$ the modular automorphism group eq. (4) takes the form

$$\sigma_s^\phi(x) = D_\phi^{is} x D_\phi^{-is}.$$

Let $(T_t)_* : L^1(\mathcal{M}) \rightarrow L^1(\mathcal{M})$ denote the pre-adjoint of the QMS T_t under the trace duality. The KMS- ϕ detailed balance condition from eq. (21) is equivalent to

$$(T_t)_*(D_\phi^{1/2} x D_\phi^{1/2}) = D_\phi^{1/2} T_t(x) D_\phi^{1/2}, \quad x \in \mathcal{M}, t \geq 0. \quad (22)$$

For $1 \leq p < \infty$ the symmetric Kosaki L^p space (Section 4.1.2) weighted by ϕ is obtained by completing $\mathcal{M}$ with respect to the norm [28], also recalled in Table 1.

$$\|\iota_{1/2}(x)\|_{p, \phi, \frac{1}{2}} = \|D_\phi^{1/(2p)} x D_\phi^{1/(2p)}\|_p, \text{ for } \|\cdot\|_p = \tau(|\cdot|^p)^{1/p}.$$

When $p = 2$, $L^2(\mathcal{M}, \phi)$ is a Hilbert space with KMS-inner product $\|\iota_{1/2}(x)\|_{2, \phi, \frac{1}{2}}^2 = \langle \Delta^{1/4} x \xi_\phi, \Delta^{1/4} x \xi_\phi \rangle$. Equation (21) shows that $(T_t)_*$ is contractive on $L^1(\mathcal{M}, \phi)$. By complex interpolation [28], it follows that it acts as a contraction on $L^p(\mathcal{M}, \phi)$ (eq. (20)) for every $1 \leq p < \infty$. We also recall from [16, Proposition 4.2] some properties of GNS-detailed-balance QMS.

Lemma 4.7. *Let $(T_t)_{t \geq 0} : \mathcal{M} \rightarrow \mathcal{M}$ be a GNS- ϕ -detailed-balance quantum Markov semigroup for a normal faithful state ϕ . Denote by $\mathcal{N}$ its fixed-point algebra (more generally, the multiplicative domain). Then*

- (i) $\mathcal{N}$ is invariant under σ_t^ϕ . Hence there exists a ϕ -preserving normal conditional expectation $E : \mathcal{M} \rightarrow \mathcal{N}$.
- (ii) $T_t|_{\mathcal{N}}$ is an involutive $*$ -automorphism satisfying

$$T_t^2 \circ E = E \circ T_t^2 = E, \quad E \circ T_t = T_t \circ E.$$

Moreover, T_t^2 is an $\mathcal{N}$ -bimodule map, $T_t^2(axb) = aT_t^2(x)b$ for all $a, b \in \mathcal{N}, x \in \mathcal{M}$.
(iii) T_t is isometric on $L_2(\mathcal{N}, \phi)$. If in addition

$$\|T_t(\text{id} - E) : L_2(\mathcal{M}, \phi) \rightarrow L_2(\mathcal{M}, \phi)\| < 1,$$

then $E = \lim_{n \rightarrow \infty} T_t^{2n} \in B(L_2(\mathcal{M}, \phi))$.

Remark 4.8 (Comparisons of GNS and KMS.). *Both GNS- and KMS-detailed balance conditions generalize the classical notion of detailed balance to QMSs. In fact, the GNS- ϕ -detailed balance condition is strictly stronger than the KMS- ϕ -detailed balance condition: as shown in [17, 50], a QMS satisfies GNS- ϕ -detailed balance if and only if (i) it satisfies KMS- ϕ -detailed balance and (ii) commutes with the modular automorphism group σ_t^ϕ .*

We will assume the GNS- ϕ detailed balance to extend the PI(p, p) (Theorem 1.2) beyond the tracial setting to a general non-tracial pair $(\mathcal{M}, \phi)$. The reason for choosing this stronger condition is its compatibility with Haagerup's reduction in Section 4.5, which allows us to reduce to tracial slices in a larger algebra. In the sequel, we shall work exclusively under that assumption.

4.3 Lindbladians, conditional expectations and gradient forms in L^p spaces

We reviewed Haagerup's $L^p(\mathcal{M})$ and Kosaki's $L^p(\mathcal{M}, \phi)_\eta$ spaces in Section 4.1, assuming that $\mathcal{M}$ is a σ -finite von Neumann algebra equipped with a normal faithful state ϕ . As established in Theorem 4.5, these L^p spaces are naturally isomorphic. In this section, we define L^p versions of Lindbladians, conditional expectation, and associated gradient forms. We will see later that compatibility with the tracial setting holds if and only if the Lindbladian satisfies the GNS- ϕ -detailed balance condition, which is the necessary regularity assumption in this framework.

Definition 4.9 (Lindbladian Generator in Haagerup's L^p Spaces). *Given the generator $L : \mathcal{M} \rightarrow \mathcal{M}$ defined in eq. (5) on $(\mathcal{M}, \phi)$ and in addition satisfy ϕ -GNS-detailed balance condition, one can define the $L^p(\mathcal{M})$ version*

$$\mathcal{L}_p^H : \text{Dom}(\mathcal{L}_p^H) \subset L^p(\mathcal{M}) \rightarrow L^p(\mathcal{M})$$

given by dense elements $h_\phi^\frac{\eta}{p} x h_\phi^\frac{1-\eta}{p} \in L^p(\mathcal{M})$ for $x \in \mathcal{M}$, $0 \leq \eta \leq 1$,

$$\mathcal{L}_p^H(h_\phi^\frac{\eta}{p} x h_\phi^\frac{1-\eta}{p}) := h_\phi^\frac{\eta}{p} L(x) h_\phi^\frac{1-\eta}{p}.$$

Definition 4.10 (Lindbladian Generator in Kosaki's L^p Spaces). *Under the assumptions of Theorem 4.9, one can define the $L^p(\mathcal{M}, \phi)_\eta$ version*

$$\mathcal{L}_p^K : \text{Dom}(\mathcal{L}_p^K) \subset L^p(\mathcal{M}, \phi)_\eta \rightarrow L^p(\mathcal{M}, \phi)_\eta$$

given by dense elements⁵ $\iota_\eta(x) := h_\phi^\eta x h_\phi^{1-\eta} \in L^p(\mathcal{M}, \phi)_\eta$, for $x \in \mathcal{M}$

$$\mathcal{L}_p^K(\iota_\eta(x)) := \iota_\eta(L(x)).$$

We denote the semigroup generated by $\mathcal{L}_p^{\{H, K\}}$ as $(T_t)_p^{\{H, K\}} := e^{-t\mathcal{L}_p^{\{H, K\}}}$. The two definitions are related via the isometric isomorphism in Equation (19); specifically,

$$\iota_{\eta, p}(\mathcal{L}_p^H(h_\phi^\frac{\eta}{p} x h_\phi^\frac{1-\eta}{p})) = \mathcal{L}_p^K(\iota_\eta(x)).$$

Based on this identification, we omit the superscripts $\{H, K\}$ and write $\mathcal{L}_p$ when the context is clear. We can also define the corresponding conditional expectations established in [25, Section 2].

Definition 4.11 (Conditional expectation in L^p spaces). *Under the assumptions of Theorem 4.1, let $\mathcal{N} \subset \mathcal{M}$ be a von Neumann subalgebra. Suppose $E_\phi : \mathcal{M} \rightarrow \mathcal{N}$ is a ϕ -preserving conditional expectation, i.e., $\phi \circ E_\phi = \phi$. Consider dense elements*

⁵Indeed, by [25, Lemma 1.1], $h_\phi^\frac{\eta}{p} \mathcal{M} h_\phi^\frac{1-\eta}{p}$ is dense in $L^p(\mathcal{M})$ for $0 < p < \infty, 0 \leq \eta \leq 1$.

$h_\phi^\frac{\eta}{p} x h_\phi^\frac{1-\eta}{p} \in L^p(\mathcal{M})$ for $x \in \mathcal{M}$, $0 \leq \eta \leq 1$, and the bimodule properties of E_ϕ , the corresponding L^p version E_p^ϕ ⁶ is:

$$E_p^\phi(h_\phi^\frac{\eta}{p} x h_\phi^\frac{1-\eta}{p}) := h_{\phi|_{\mathcal{N}}}^\frac{\eta}{p} E^\phi(x) h_{\phi|_{\mathcal{N}}}^\frac{1-\eta}{p} \quad (23)$$

It is also known that $L^p(\mathcal{N}) = L^p(\mathcal{N}, \phi|_{\mathcal{N}})$ can be naturally isometrically identified with a subspace of $L^p(\mathcal{M})$. From now on we identify ϕ , ι_η , σ_t , and τ (the canonical trace from eq. (18)) with their restrictions on $\mathcal{N}$, and do not distinguish them unless needed for clarity. Under these identifications, we summarize the compatible QMSs in Figure 3.

$$\begin{array}{ccccc} \mathcal{M} & \xrightarrow{\iota_\eta} & L^1(\mathcal{M}) & \longleftrightarrow & L^p(\mathcal{M}, \phi)_\eta \cong L^p(\mathcal{M}) \\ \downarrow T_t & & \downarrow (T_t)_* & & \downarrow (T_t)_p \\ \mathcal{M} & \xrightarrow{\iota_\eta} & L^1(\mathcal{M}) & \longleftrightarrow & L^p(\mathcal{M}, \phi)_\eta \cong L^p(\mathcal{M}) \\ \downarrow E & & \downarrow E_1 & & \downarrow E_p^\phi \\ \mathcal{N} & \xrightarrow{\iota_\eta} & L^1(\mathcal{N}) & \longleftrightarrow & L^p(\mathcal{N}, \phi)_\eta \cong L^p(\mathcal{N}) \end{array}$$

Fig. 3 Kosaki's L^p -interpolation spaces for $\mathcal{M}$ and their compatible QMSs, via the functoriality of complex interpolation [7].

Lemma 4.12 ([25, Lemma 2.2]). *Under the assumptions of Theorem 4.11, for each $1 \leq p < \infty$, E_p^ϕ extends to a contractive projection, hence isometric on its range.*

In particular, in Haagerup's L^p spaces

$$\|E_p^\phi(x)\|_{L^p(\mathcal{M})} \leq \|x\|_{L^p(\mathcal{N})}, \quad x \in L^p(\mathcal{M}),$$

with equality for $x \in L^p(\mathcal{N})$. Equivalently, in Kosaki's L^p spaces,

$$\|\iota_{\eta,p}^\phi(E^\phi(x))\|_{p,\phi,\eta} = \|E_1^\phi(\iota_{\eta,p}^\phi(x))\|_{p,\phi,\eta} \leq \|\iota_\eta^\phi|_{\mathcal{N}}(x)\|_{p,\phi|_{\mathcal{N}},\eta}, \quad x \in \mathcal{M},$$

with equality for $x \in \mathcal{N}$.

We next explore the compatible definitions for gradient forms in both L^p spaces and their identification.

Definition 4.13 (Gradient form in Haagerup's L^p spaces). *Let*

$$\mathfrak{D}_p := \{(x, y) : x, y \in \text{Dom}(\mathcal{L}_p), \ x^*y \in \text{Dom}(\mathcal{L}_{p/2})\}.$$

⁶By [25, Lemma 2.1] for any $0 \leq \eta_1, \eta_2 \leq 1$, the L^p conditional expectation are identical, thus E_p^ϕ have no η dependence.

The Haagerup L^p gradient form for the Lindblad $\mathcal{L}_p$ is the sesquilinear map

$$\Gamma_p : \mathfrak{D}_p \longrightarrow L^{p/2}(\mathcal{M})$$

defined by

$$\Gamma_p(x, y) = \frac{1}{2} (\mathcal{L}_p(x^*)y + x^* \mathcal{L}_p(y) - \mathcal{L}_{p/2}(x^*y)). \quad (24)$$

For $(a, a) \in \mathfrak{D}_p$, we use the shorthand

$$\|a\|_{\Gamma_p} := \|\Gamma_p(a, a)\|_{L^{p/2}(\mathcal{M})}^{1/2}.$$

The same convention applies when the L^p gradient form carries additional decorations, such as $\Gamma_{L^\varepsilon, p}$ or Γ_p^k .

Definition 4.14 (Gradient form in Kosaki's L^p spaces). For $x, y \in \mathcal{M}_a^\sigma \cap \text{Dom}(L)$, define the gradient form compatible with the Kosaki interpolation space by

$$\Gamma_\eta^{(p)}(x, y) := \Gamma(\sigma_{-\frac{i\eta}{p}}(x), \sigma_{-\frac{i\eta}{p}}(y)),$$

Here $\mathcal{M}_a^\sigma$ is the set of all entire elements of $\mathcal{M}$ ⁷, $\sigma_{-\frac{i\eta}{p}}(x) = h_\phi^{\frac{\eta}{p}} x h_\phi^{-\frac{\eta}{p}}$ is the modular automorphism at complex time $t = \frac{-i\eta}{p}$, and Γ is defined in eq. (7). For such x , we use the shorthand

$$\|x\|_{\Gamma_\eta^{(p)}} := \|\iota_{\frac{1}{2}}(\Gamma_\eta^{(p)}(x, x))\|_{\frac{p}{2}, \phi, \frac{1}{2}}^{1/2}.$$

The identification between two gradient forms is for dense elements,

$$\Gamma_p(h_\phi^{\frac{\eta}{p}} x h_\phi^{\frac{1-\eta}{p}}, h_\phi^{\frac{\eta}{p}} y h_\phi^{\frac{1-\eta}{p}}) = h_\phi^{\frac{1}{p}} \Gamma_\eta^{(p)}(x, y) h_\phi^{\frac{1}{p}}, \quad x, y \in \mathcal{M}$$

In the tracial case, one can replace h_ϕ by D_ϕ , where D_ϕ is the density of ϕ with respect to the trace, i.e., $\phi(\cdot) = \tau(D_\phi \cdot)$.

4.4 Translation towards non-trace-symmetric QMS

In this section, we rewrite Theorem 1.2 and Theorem 3.9 for τ -symmetric QMS in $(\mathcal{M}, \tau)$ in terms of relevant L^p quantities as in section 4.3 with an additional assumption of the KMS- ϕ -detailed balance condition. The state ϕ admits a density D_ϕ with respect to the trace τ , satisfying $\phi(\cdot) = \tau(D_\phi \cdot)$. The following results follow directly from rewriting and the factorization property in eq. (22).

Proposition 4.15 (Haagerup's PI(p, p)). *Given the same conditions as the theorem 1.2 and assuming further that the QMS $(T_t)_{t \geq 0}$ is KMS- ϕ -detailed balance, the following inequality holds in Haagerup's L^p spaces for all $p = 2$ and $p \geq 3$, and for all*

⁷An element $x \in \mathcal{M}$ is said to be entire if the function $\mathbb{R} \ni t \mapsto \sigma_t(x) \in \mathcal{M}$ can be extended to an $\mathcal{M}$ -valued entire function over $\mathbb{C}$. It follows from [47, Lemma 2.3] that $\mathcal{M}_a^\sigma$ is a σ -weakly dense $*$ -subalgebra of $\mathcal{M}$.

dense $x \in \mathcal{M}_{sa}$, $a := D_\phi^{\frac{1}{2p}} x D_\phi^{\frac{1}{2p}} \in L^p(\mathcal{M})_{sa}$ in the domain:

$$\|a - E_p^\phi(a)\|_{L^p(\mathcal{M})} \leq \frac{p}{\sqrt{2\alpha}} \|a\|_{\Gamma_p}.$$

Furthermore, for $a \in L^p(\mathcal{M})$ in the domain,

$$\|a - E_p^\phi(a)\|_{L^p(\mathcal{M})} \leq \frac{p}{\sqrt{2\alpha}} (\|a\|_{\Gamma_p} + \|a^*\|_{\Gamma_p}).$$

Proposition 4.16 (Kosaki's PI(p, p)). *Given the same conditions as in Theorem 4.15, for every $p = 2$ and $p \geq 3$ and $x \in \mathcal{M}_{sa}$ in the domain, we have the inequality in Kosaki's L^p spaces:*

$$\|\iota_{\frac{1}{2}}(x - E(x))\|_{p, \phi, \frac{1}{2}} \leq \frac{p}{\sqrt{2\alpha}} \|x\|_{\Gamma_{\frac{1}{2}}^{(p)}}.$$

Furthermore, for $x \in \mathcal{M}$ in the domain,

$$\|\iota_\eta(x - E(x))\|_{p, \phi, \eta} \leq \frac{p}{\sqrt{2\alpha}} (\|x\|_{\Gamma_\eta^{(p)}} + \|x^*\|_{\Gamma_{1-\eta}^{(p)}}).$$

In particular, the one-sided version is

$$\|(x - E(x))D_\phi\|_{p, \phi, 0} \leq \frac{p}{\sqrt{2\alpha}} (\|x\|_{\Gamma_0^{(p)}} + \|x^*\|_{\Gamma_1^{(p)}}), \quad (25)$$

Moreover, Theorems 4.15 and 4.16 remain well-defined for non-tracial von Neumann algebras: one needs only replace the canonical density D_ϕ by h_ϕ (cf. item (v) of Theorem 4.2). The replaced h_ϕ is defined in the absence of a trace.

As discussed in Theorem 4.8, when we have a σ -finite $(\mathcal{M}, \phi)$, the τ -symmetric condition is no longer available. Therefore, the natural replacement is the stronger GNS- ϕ -detailed balance. We formulate our main theorem below and prove it in later sections.

Theorem 4.17 (PI(p, p) for GNS- ϕ -detailed balance). *Let $(\mathcal{M}, \phi)$ be a σ -finite von Neumann algebra equipped with a faithful normal state ϕ , and let $(T_t)_{t \geq 0}$ be a GNS- ϕ -detailed-balance QMS with fixed-point algebra $\mathcal{N}$. Suppose $(T_t)_{t \geq 0}$ has a spectral gap*

$$\alpha := \inf_{\substack{x \in \ker E \\ x \neq 0}} \frac{\phi(x^* L(x))}{\phi(x^* x)} > 0.$$

Let E_p^ϕ be the induced conditional expectation from $L^p(\mathcal{M})$ onto $L^p(\mathcal{N})$. Then, for $p = 2$ or $p \geq 3$ and $a \in L^p(\mathcal{M})_{sa}$ in the domain, the following holds in the Haagerup L^p -space:

$$\|a - E_p^\phi(a)\|_{L^p(\mathcal{M})} \leq \frac{p}{\sqrt{2\alpha}} \|a\|_{\Gamma_p}.$$

Equivalently, using symmetric Kosaki's L^p spaces notation, for $x \in \mathcal{M}_{sa}$ in the domain

$$\|\iota_{\frac{1}{2}}(x - E^\phi(x))\|_{p, \phi, \frac{1}{2}} \leq \frac{p}{\sqrt{2\alpha}} \|x\|_{\Gamma_{\frac{1}{2}}^{(p)}}.$$

Corollary 4.18 (GNS- ϕ -detailed balance PI(p, p) for non-self-adjoint elements). *Under the construction of Theorem 4.17, for $a \in L^p(\mathcal{M})$ in the domain,*

$$\|a - E_p^\phi(a)\|_{L^p(\mathcal{M})} \leq \frac{p}{\sqrt{2\alpha}} (\|a\|_{\Gamma_p} + \|a^*\|_{\Gamma_p}).$$

Equivalently, using Kosaki's L^p spaces notation, for $\eta \in [0, 1]$, $x \in \mathcal{M}$ in the domain

$$\|\iota_\eta(x - E(x))\|_{p, \phi, \eta} \leq \frac{p}{\sqrt{2\alpha}} (\|x\|_{\Gamma_\eta^{(p)}} + \|x^*\|_{\Gamma_{1-\eta}^{(p)}}).$$

Proof. As in Theorem 3.9, write $a = \operatorname{Re}(a) + i\operatorname{Im}(a)$ and apply the bound to $\operatorname{Re}(a)$ and $\operatorname{Im}(a)$. $\square$

Remark 4.19 (Domain convention). *In the tracial setting, Cipriani and Sauvageot [11] identified and extended the domain of the gradient form. Indeed, they showed that $\operatorname{Dom}(L^{1/2})$ is a $*$ -algebra by constructing a derivation such that*

$$\operatorname{Dom}(\bar{\delta}) = \operatorname{Dom}(L^{1/2}), \quad \|\bar{\delta}(x)\|_{\mathcal{H}}^2 = \|L^{1/2}x\|_2^2 = \|\Gamma(x, x)\|_1.$$

Moreover, if the state is tracial, then $T_t(x) \in \operatorname{Dom}(L^{1/2}) \cap \operatorname{Dom}(L)$ for every $x \in \mathcal{M}$ and $t > 0$. Thus, $x_n = T_{1/n}(x)$ gives an approximating sequence. Although the gradient seminorm is initially defined only on a regular core, one would like to define a gradient seminorm for general elements of $L^p(\mathcal{M})$ by approximation. For a general $a \in L^p(\mathcal{M})$, the gradient seminorm in Theorem 4.17 is therefore understood via analytic approximation

$$\|a\|_{\Gamma_p} := \inf_{\substack{a_n \in \mathcal{C}_p \\ \|a_n - a\|_{L^p(\mathcal{M})} \rightarrow 0}} \liminf_{n \rightarrow \infty} \|a_n\|_{\Gamma_p}, \quad (26)$$

where

$$\mathcal{C}_p := \left\{ h_\phi^{1/(2p)} x h_\phi^{1/(2p)} : x \in \mathcal{M}_a^\sigma \cap \operatorname{Dom}(L) \right\},$$

and the infimum is $+\infty$ if no such approximating sequence exists. We defer a detailed treatment of the domain issue to Appendix D.

4.5 Haagerup's reduction

To generalize our results to the non-tracial case, we introduce Haagerup's reduction [19], which shows that every von Neumann algebra admits finite approximations. We later show that the tracial PI(p, p) inequality can be transferred to a general von Neumann algebra via Haagerup reduction. Throughout this section, let $G = \bigcup_{n \geq 1} 2^{-n}\mathbb{Z} \subset \mathbb{R}$ and let $\mathcal{M}$ be a σ -finite⁸ von Neumann algebra equipped with a normal faithful

⁸Mutually orthogonal projections in $\mathcal{M}$ are at most countable.

state ϕ . We denote by σ_t^ϕ the modular automorphism group of ϕ on $\mathcal{M}$, and form the crossed-product $\widehat{\mathcal{M}} := \mathcal{M} \rtimes_{\sigma^\phi} G$. There is a canonical normal conditional expectation $\mathcal{E} : \widehat{\mathcal{M}} \rightarrow \mathcal{M}$ ⁹. Finally, let $\widehat{\phi} = \phi \circ \mathcal{E}$ be the dual weight on $\widehat{\mathcal{M}}$, which is also a normal faithful state.

Theorem 4.20 (Haagerup reduction [19, Theorem 2.1]). *Following the notation above, there exists an increasing sequence $(\mathcal{M}_k)_{k \geq 1}$ of von Neumann subalgebras of $\widehat{\mathcal{M}}$ satisfying:*

- (i) *each $\mathcal{M}_k$ is finite with normal faithful trace τ_k ;*
- (ii) *$\bigcup_{k \geq 1} \mathcal{M}_k$ is w^* -dense in $\widehat{\mathcal{M}}$;*
- (iii) *there is a family of normal faithful conditional expectation $(\mathcal{E}_k)_{k \geq 1} : \widehat{\mathcal{M}} \rightarrow \mathcal{M}_k$ such that*

$$\widehat{\phi} \circ \mathcal{E}_k = \widehat{\phi}, \quad \sigma_t^{\widehat{\phi}} \circ \mathcal{E}_k = \mathcal{E}_k \circ \sigma_t^{\widehat{\phi}}, \quad t \in \mathbb{R};$$

- (iv) *let $\phi_k(\cdot) := \widehat{\phi}|_{\mathcal{M}_k}(\cdot) = \tau_k(d_k \cdot)$ for $d_k \in L(G) \subset \mathcal{Z}(\widehat{\mathcal{M}}_{\widehat{\phi}})$ so that $\sigma_t^{\widehat{\phi}}(d_k) = d_k$; Furthermore $a_k \leq d_k \leq a_k^{-1}$ for some $a_k \in \mathbb{R}^+$;*
- (v) *$\lim_{k \rightarrow \infty} \mathcal{E}_k(x) = x$ in σ -strong topology for any $x \in \widehat{\mathcal{M}}$.*
- (vi) ([16, Section 4.2]) *$\varphi_k \rightarrow \widehat{\varphi}$ in the weak topology on $S(\mathcal{M})$, for $\varphi \in S(\mathcal{M})$ and $\varphi_k := \widehat{\varphi}|_{\mathcal{M}_k}$.*

We apply Haagerup's reduction Theorem 4.20 to the GNS- ϕ -detailed-balance QMS $(T_t)_{t \geq 0}$. We call $(\widehat{T}_t)_{t \geq 0}$ its canonical extension to $\widehat{\mathcal{M}}$, defined by $\widehat{T}_t := T_t \otimes \text{Id}_{B(l_2(G))}|_{\widehat{\mathcal{M}}}$. It is also shown in [16, Section 4.2] to be a GNS- $\widehat{\phi}$ -detailed-balance QMS with fixed-point algebra $\widehat{\mathcal{N}} := \mathcal{N} \rtimes_{\sigma^\phi} G$.

Moreover, the $\widehat{\phi}$ -preserving conditional expectation $\widehat{E} : \widehat{\mathcal{M}} \rightarrow \widehat{\mathcal{N}}$ is just the canonical extension of $E : \mathcal{M} \rightarrow \mathcal{N}$. It turns out that $(\widehat{T}_t)_{t \geq 0}$ is compatible with the finite approximations in Haagerup reduction, summarized in the theorem below.

Theorem 4.21 ([16, Lemma 4.5]). *Let $(T_t)_{t \geq 0} : \mathcal{M} \rightarrow \mathcal{M}$ be a GNS- ϕ -symmetric quantum Markov semigroup. With the notation above and $t \geq 0$, the following hold:*

- (i) *For each k and $s \in \mathbb{R}$,*

$$\widehat{T}_t \circ \sigma_s^{\phi_k} = \sigma_s^{\phi_k} \circ \widehat{T}_t.$$

Consequently, $\widehat{T}_t$ commutes with $\mathcal{E}$, $\widehat{\mathcal{E}}$, and $\mathcal{E}_k$. In particular, $\widehat{T}_t(\mathcal{M}_k) \subseteq \mathcal{M}_k$.

- (ii) *The restriction*

$$T_{t,k} := \widehat{T}_t|_{\mathcal{M}_k} : \mathcal{M}_k \longrightarrow \mathcal{M}_k$$

is a τ_k -symmetric normal unital completely positive map.

- (iii) *Let $\mathcal{N}_k := \mathcal{M}_k \cap \widehat{\mathcal{N}} \subseteq \mathcal{M}_k$ be the fixed-point algebra of $T_{t,k}$. Then*

$$E_k := \widehat{E}|_{\mathcal{M}_k} : \mathcal{M}_k \longrightarrow \mathcal{N}_k$$

is the τ_k -preserving conditional expectation onto $\mathcal{N}_k$.

⁹We avoid using the crossed product $\mathcal{M} \rtimes \mathbb{R}$ because, by [46, Sec. IX.4], the existence of a normal conditional expectation onto $\mathcal{M}$ forces the factor type to be preserved. In particular, even if $\mathcal{M}$ is type III, its crossed product with $\mathbb{R}$ is always semifinite (type II_∞) due to the presence of a faithful trace.

Combining the preceding results with [19, Eq. (6.4)], which gives

$$\widehat{E} \circ \mathcal{E} = \mathcal{E} \circ \widehat{E} \quad \text{and} \quad \widehat{E} \circ \mathcal{E}_k = \mathcal{E}_k \circ \widehat{E},$$

we obtain the following commuting diagrams Figure 4.

$$\begin{array}{ccccc} \mathcal{M} & \xleftarrow{\mathcal{E}} & \widehat{\mathcal{M}} & \xrightarrow{\mathcal{E}_k} & \mathcal{M}_k \\ \downarrow T_t & & \downarrow \widehat{T}_t & & \downarrow T_{t,k} \\ \mathcal{M} & \xleftarrow{\mathcal{E}} & \widehat{\mathcal{M}} & \xrightarrow{\mathcal{E}_k} & \mathcal{M}_k \\ \downarrow E & & \downarrow \widehat{E} & & \downarrow E_k \\ \mathcal{N} & \xleftarrow{\mathcal{E}} & \widehat{\mathcal{N}} & \xrightarrow{\mathcal{E}_k} & \mathcal{N}_k \end{array}$$

Fig. 4 Haagerup reduction commuting diagram: the quantum Markov semigroup T_t on $\mathcal{M}$, its finite approximants $\mathcal{M}_k$, and the conditional expectations E, E_k down to $\mathcal{N}$.

We have the following corollary on the tracial slices $(\mathcal{M}_k, \tau_k)$:

Corollary 4.22. *Using the notation above, for every $\alpha \in \mathbb{R}$ and every $x \in \mathcal{M}_k$,*

$$T_{t,k}(d_k^\alpha x) = d_k^\alpha T_{t,k}(x),$$

and similarly

$$T_{t,k}(x d_k^\alpha) = T_{t,k}(x) d_k^\alpha.$$

Proof. Since $d_k \in \mathcal{M}_k$ and $T_{t,k} = \widehat{T}_t|_{\mathcal{M}_k}$,

$$T_{t,k}(d_k) = \widehat{T}_t(d_k) = d_k.$$

Because $T_{t,k}$ is unital completely positive and τ_k -preserving, Kadison–Schwarz and faithfulness of τ_k imply

$$T_{t,k}(d_k^2) = T_{t,k}(d_k)^2 = d_k^2.$$

Thus d_k lies in the multiplicative domain of $T_{t,k}$. By functional calculus, the same holds for d_k^α , and $T_{t,k}(d_k^\alpha) = d_k^\alpha$. Hence

$$T_{t,k}(d_k^\alpha x) = T_{t,k}(d_k^\alpha) T_{t,k}(x) = d_k^\alpha T_{t,k}(x).$$

The right multiplication identity is identical. □

4.6 Proof of L^p version of PI(p, p)

We are now ready to prove our main theorem Theorem 4.17. Unless otherwise stated, we use the notation established in previous sections. The outline is to prove that

PI(p, p) holds on the tracial slices $(\mathcal{M}_k, \tau_k)$ for all $k \geq 1$ and extend it to $(\mathcal{M}, \phi)$ by Haagerup's reduction.

Proof. We first verify that the spectral gap α is inherited by $T_{t,k}$. Combining this with (ii) of Theorem 4.21, it follows from Theorem 1.2 that PI(p, p) holds on each slice $(\mathcal{M}_k, \tau_k)$. Indeed, given $\alpha > 0$ spectral gap for T_t , we have for $x \in \mathcal{M}$,

$$\|T_t(x) - E(x)\|_{L^2(\mathcal{M})} \leq e^{-\alpha t} \|x - E(x)\|_{L^2(\mathcal{M})}.$$

Recall that $\widehat{T}_t := T_t \otimes \text{Id}_{B(\ell_2(G))} |_{\widehat{\mathcal{M}}}$. By direct calculation, or equivalently by Theorem B.1, using that the spectral gap of Id is $\alpha_{\text{Id}} = \infty$, we see that $\widehat{T}_t$ has the same spectral gap α as T_t . By isometry of norms for $\mathcal{E}_k$ on L^2 spaces using Lemma 4.12, it follows that $\widehat{T}_t$ has the same spectral gap α with respect to $\phi_k(\cdot) = \tau_k(d_k \cdot)$ when restricted to each slice $(\mathcal{M}_k, \tau_k)$ for $k \geq 1$. Define

$$U_k : L^2(\mathcal{M}_k, \phi_k) \rightarrow L^2(\mathcal{M}_k, \tau_k), \quad U_k(x) = x d_k^{1/2}.$$

By (iv) of Theorem 4.20, d_k is boundedly invertible, so U_k is unitary. Moreover, since $d_k^{1/2} \in \mathcal{N}_k$, Theorem 4.22 and the $\mathcal{N}_k$ -bimodularity of E_k imply

$$U_k T_{t,k} = T_{t,k} U_k, \quad U_k E_k = E_k U_k.$$

Therefore $(T_{t,k}, E_k)$ on $L^2(\mathcal{M}_k, \phi_k)$ is unitarily equivalent to $(T_{t,k}, E_k)$ on $L^2(\mathcal{M}_k, \tau_k)$. In particular, $T_{t,k}$ has the same spectral gap α with respect to τ_k .

In the sequel, we adopt the convention that all objects associated with the restricted semigroup $T_{t,k}$ carry the subscript k . In particular, we set

$$L_k := \widehat{L}|_{\mathcal{M}_k}, \quad \mathcal{L}_{p,k}^K := \widehat{\mathcal{L}}_p^K |_{\mathcal{M}_k}, \quad \Gamma_\eta^{(p,k)} := \widehat{\Gamma}_\eta^{(p)} |_{\mathcal{M}_k \times \mathcal{M}_k},$$

so that L_k , $\mathcal{L}_{p,k}^K$ and $\Gamma_\eta^{(p,k)}$ are the Lindbladian, the Kosaki's L^p Lindbladian and L^p gradient form of $T_{t,k}$ on $\mathcal{M}_k$. Given $k \geq 1$, let $x_k \in (\mathcal{M}_k)_{sa} \cap \widehat{\mathcal{M}}_a^\sigma \cap \text{Dom}(L_k)$, where $\widehat{\mathcal{M}}_a^\sigma$ the set of all entire elements of $\widehat{\mathcal{M}}$, and d_k the density of $\phi_k := \widehat{\phi}|_{\mathcal{M}_k}$ under trace τ_k . By (iv) of Theorem 4.20, $d_k \in L(G) \subset \widehat{\mathcal{N}} := \mathcal{N} \rtimes_{\sigma^\phi} G$ and $d_k \in \mathcal{M}_k$, it follows that $d_k \in \mathcal{N}_k = \mathcal{M}_k \cap \widehat{\mathcal{N}}$. We can assert the following

$$\mathcal{L}_{p,k}^K(d_k^{\frac{1}{2}} x_k d_k^{\frac{1}{2}}) = d_k^{\frac{1}{2}} L_k(x_k) d_k^{\frac{1}{2}},$$

which agrees with Theorem 4.9. Thus, we can write the PI(p, p) in Kosaki's L^p spaces as in Theorem 4.16. For all $p = 2$ and $p \geq 3$, and let $\iota_\eta(x) = h_\phi^\eta x h_\phi^{(1-\eta)}$ ¹⁰ for $\eta \in [0, 1]$,

¹⁰We omit the ϕ -subscript on ι whenever the norm's subscript makes the state clear. It is also known that $L^p(\mathcal{N})$ is naturally isometrically embedded in $L^p(\mathcal{M})$ [25] whenever there exists a conditional expectation $E : \mathcal{M} \rightarrow \mathcal{N}$, so no further distinction is needed.

$$\|\iota_{\frac{1}{2}}(x_k - E_k(x_k))\|_{p, \phi_k, \frac{1}{2}} \leq \frac{p}{\sqrt{2\alpha}} \|\iota_{\frac{1}{2}}(\Gamma_{\frac{1}{2}}^{(p,k)}(x_k, x_k))\|_{\frac{p}{2}, \phi_k, \frac{1}{2}}^{1/2}.$$

As we are working with self-adjoint elements we denote $\|\cdot\|_{p, \phi, \frac{1}{2}}$ just as $\|\cdot\|_{p, \phi}$ in the following. Taking the limit $k \rightarrow \infty$, the above relations remain valid, and hence $\text{PI}(p, p)$ holds in $\widehat{\mathcal{M}}$ and thus in $\mathcal{M}$.

To see this, for $x \in \mathcal{B} := \mathcal{M}_{sa} \cap \mathcal{M}_a^\sigma \cap \text{Dom}(L) \subset \mathcal{M} \subset \widehat{\mathcal{M}}$, we have $x_k := \mathcal{E}_k(x) \in (\mathcal{M}_k)_{sa}$, $\mathcal{E}_{k,1} \circ \iota_\eta = \iota_\eta \circ \mathcal{E}_k$ and $E_k \circ \mathcal{E}_k = \mathcal{E}_k \circ \widehat{E} = \mathcal{E}_k \circ E^{11}$ from Figures 3 and 4, the left-hand side rewrites as

$$\|\iota_{\frac{1}{2}}(x_k - E_k(x_k))\|_{p, \phi_k} = \|\iota_{\frac{1}{2}}(\mathcal{E}_k(x) - E_k(\mathcal{E}_k(x)))\|_{p, \phi_k} = \|\mathcal{E}_{k,1}(\iota_{\frac{1}{2}}(x - E(x)))\|_{p, \phi_k}. \quad (27)$$

Taking the limit $k \rightarrow \infty$ and applying (v) of Theorem 4.20, $\lim_{k \rightarrow \infty} \mathcal{E}_k(x) = x$ in σ -strong topology for any $x \in \mathcal{B} \subset \widehat{\mathcal{M}}$ give us

$$\lim_{k \rightarrow \infty} \|\mathcal{E}_{k,1}(\iota_{\frac{1}{2}}(x - E(x)))\|_{p, \phi_k} = \|\iota_{\frac{1}{2}}(x - E(x))\|_{p, \widehat{\phi}} = \|\iota_{\frac{1}{2}}(x - E(x))\|_{p, \phi}, \quad (28)$$

where the last equality uses Lemma 4.12 where the L^p extensions of $\mathcal{E} : \widehat{\mathcal{M}} \rightarrow \mathcal{M}$ is a contractive projection.

Similarly, for the RHS, we can remove one conditional expectation as in eqs. (27) and (28),

$$\|\iota_{\frac{1}{2}}(\Gamma_{\frac{1}{2}}^{(p,k)}(x_k, x_k))\|_{\frac{p}{2}, \phi_k}^{1/2} \leq \|\iota_{\frac{1}{2}}(\widehat{\Gamma}_{\frac{1}{2}}^{(p)}(x, \mathcal{E}_k(x)))\|_{\frac{p}{2}, \widehat{\phi}}^{1/2}$$

We know $\widehat{\Gamma}_{\frac{1}{2}}^{(p)}(x, \mathcal{E}_k(x))$ is linear on $\mathcal{E}_k(x)$ from Theorem 4.14,

$$\widehat{\Gamma}_{\frac{1}{2}}^{(p)}(x, \mathcal{E}_k(x)) = \Gamma(\sigma_{-\frac{i}{2p}}^{\widehat{\phi}}(x), \sigma_{-\frac{i}{2p}}^{\widehat{\phi}}(\mathcal{E}_k(x))) = \Gamma(\sigma_{-\frac{i}{2p}}^{\widehat{\phi}}(x), \mathcal{E}_k(\sigma_{-\frac{i}{2p}}^{\widehat{\phi}}(x))),$$

where the second equality is due to $\sigma_t^{\widehat{\phi}} \circ \mathcal{E}_k = \mathcal{E}_k \circ \sigma_t^{\widehat{\phi}}$ by (iii) of Theorem 4.20. Let $y := \sigma_{-\frac{i}{2p}}^{\widehat{\phi}}(x) = \sigma_{-\frac{i}{2p}}^{\widehat{\phi}}|_{\mathcal{M}}(x) = \sigma_{-\frac{i}{2p}}^{\phi}(x) \in \mathcal{M}$ for $x \in \mathcal{M}_a^\sigma$, and $y_k := \mathcal{E}_k(y)$. We aim to prove that

$$\lim_{k \rightarrow \infty} \|\iota_{\frac{1}{2}}(\widehat{\Gamma}(y, y_k))\|_{\frac{p}{2}, \widehat{\phi}, \frac{1}{2}}^{1/2} \leq \|\iota_{\frac{1}{2}}(\widehat{\Gamma}(y, y))\|_{\frac{p}{2}, \widehat{\phi}, \frac{1}{2}}^{1/2} = \|\iota_{\frac{1}{2}}(\Gamma(y, y))\|_{\frac{p}{2}, \phi, \frac{1}{2}}^{1/2}, \quad (29)$$

where the last equality is due to the isometry of the norm from Lemma 4.12. There are technical complications due to the domain issue for the gradient form $\Gamma : \text{Dom}(L)$ is not closed under multiplication. In particular, for $y, y_k \in \text{Dom}(L)$ one may have $y^* y_k \notin \text{Dom}(L) \subset \text{Dom}(\widehat{L})$. To work around this issue, we employ the ϵ -regularization, so that all expressions are well defined and the domain problem does not obstruct the

¹¹Since $\widehat{E}$ is the canonical extension of E , we write E for $\widehat{E}$ when acting on $x \in \mathcal{M}$.

passage to the limit. Recall in this framework, for all $x, y \in \mathcal{A} := \text{Dom}(L) \cap \mathcal{M} \subset \text{Dom}(\widehat{L}) \cap \widehat{\mathcal{M}}$,

$$\widehat{\Gamma}_\epsilon(x, y) := \frac{1}{2} \left(x^* \widehat{L}_\epsilon(y) + \widehat{L}_\epsilon(x)^* y - \widehat{L}_\epsilon(x^* y) \right), \text{ where } \widehat{L}_\epsilon := \frac{\widehat{L}}{1 + \epsilon \widehat{L}}. \quad (30)$$

It follows from [11, Prop. 2.5] that, for $\epsilon > 0$, the resolvent maps $(I + \epsilon \widehat{L})^{-1}$ are bounded, completely positive contractions on $L^2(\widehat{\mathcal{M}})$ and the regulated operators $\widehat{L}_\epsilon$ are completely bounded endomorphisms of $\widehat{\mathcal{M}}$.

These properties guarantee that the regularized gradient form above is well defined and uniformly bounded (Theorem C.4). As shown in Appendix C, detailed balance for L passes to the regulator L_ϵ , so that the tracial PI(p, p) inequality holds for the regularization $L_{k, \epsilon} = L_k(I + \epsilon L_k)^{-1}$ on $\mathcal{M}_k$, and the spectral gap satisfies $\alpha_\epsilon = \alpha/(1 + \epsilon\alpha)$.

Meanwhile, we restrict to the simplified case in which we assume that $\mathcal{A}$ is a weak*-dense $*$ -algebra, given by $\mathcal{A} = \text{Dom}(L) \cap \mathcal{M}$. By Theorem C.3, $\widehat{\Gamma}_\epsilon(x, y) \rightarrow \widehat{\Gamma}(x, y)$ strongly for all $x, y \in \mathcal{A}$. To prove eq. (29), we first work with the regularized operators $\widehat{\Gamma}_\epsilon$. By (v) of Theorem 4.20, we have $\lim_{k \rightarrow \infty} y_k = y$ in the σ -strong topology. Since $\widehat{L}_\epsilon = \widehat{L}(1 + \epsilon \widehat{L})^{-1}$ is bounded for each $\epsilon > 0$, it is straightforward to check that

$$\lim_{k \rightarrow \infty} \widehat{\Gamma}_\epsilon(y, y_k) = \lim_{k \rightarrow \infty} \frac{1}{2} \left(\widehat{L}_\epsilon(y^*) y_k + y^* \widehat{L}_\epsilon(y_k) - \widehat{L}_\epsilon(y^* y_k) \right) = \widehat{\Gamma}_\epsilon(y, y) \quad (31)$$

strongly. After taking $\epsilon \downarrow 0$ and applying Lemma C.2, we obtain the equality form of eq. (29) in the $\frac{p}{2}$ -norm. For the general case when there is no $*$ -algebra $\mathcal{A}$, we use the weak* convergence definition extended by [11, Lemma 9.1] (eq. (8)).

$$\Gamma(x, y) := \text{w}^*\text{-}\lim_{\epsilon \downarrow 0} \Gamma_\epsilon(x, y).$$

For an arbitrary self-adjoint $x \in \mathcal{M}_{sa}$, we consider a family of analytic approximations $\psi_m(x) \in \mathcal{M}_a^\sigma$. Recall from [47, Chap. VIII.2.3] that these approximations may be defined by

$$\psi_m(x) := \int_{\mathbb{R}} \sigma_t^\phi(x) K_m(t) dt,$$

where K_m is a family of suitable Gaussian kernels. Each map ψ_m commutes with the GNS-detailed-balance semigroup $\{T_t\}_{t \geq 0}$ because the semigroup commutes with the modular automorphism group $\{\sigma_t^\phi\}_{t \in \mathbb{R}}$. Consequently, $T_{1/n}(\psi_m(x)) \in \mathcal{B}$. Moreover, $T_{1/n}(\psi_m(x)) \rightarrow x$ strongly as $n, m \rightarrow \infty$. We apply the PI(p, p) inequality to these approximating elements and pass to the limit on both sides, interpreting the Γ_p -norm from Equation (26).

By the density of $\bigcup_k \mathcal{M}_k$, stated in part (ii) of Theorem 4.20, the above arguments establish the L^p version of PI(p, p) on $(\mathcal{M}, \phi)$ for every $x \in \mathcal{M}_{sa} \cap \text{Dom}(L)$.

Equivalently, in Haagerup's functorial formulation of L^p -spaces, for every $a = h_\phi^{1/(2p)} x h_\phi^{1/(2p)} \in L^p(\mathcal{M})_{sa}$ with $x \in \mathcal{M}_{sa} \cap \text{Dom}(L)$, we have

$$\|a - E_p^\phi(a)\|_{L^p(\mathcal{M})} \leq \frac{p}{\sqrt{2\alpha}} \|\Gamma_p(a, a)\|_{L^{p/2}(\mathcal{M})}^{1/2}.$$

Since Haagerup's functorial construction is independent of any particular choice of trace, this L^p inequality is canonically well-defined for all $a \in L^p(\mathcal{M})_{sa}$. We defer the details of the general case to Appendix D. $\square$

5 Examples and Applications

5.1 Khintchine inequalities with Rademacher random variables

We present a degree-one example in which the $\text{PI}(p, p)$ inequality yields Khintchine-type estimates. This example also shows that, in the non-self-adjoint extension $\text{PI}(p, p)$ from Theorem 4.18, the second term (the $\|a^*\|_{\Gamma_p}$ contribution) is genuinely necessary and cannot be removed.

Let $\Omega = \{-1, 1\}^n$ be equipped with the uniform probability measure, and let $\varepsilon_1, \dots, \varepsilon_n$ denote the coordinate (Rademacher) functions. We consider the von Neumann algebra

$$\mathcal{A} := L_\infty(\Omega) \bar{\otimes} \mathcal{M} \cong L_\infty(\Omega, \mathcal{M}),$$

equipped with the product state $\psi := \mathbb{E} \otimes \phi$, where $\mathbb{E}$ is the uniform expectation on $L_\infty(\Omega)$. Concretely, elements of $\mathcal{A}$ are bounded $\mathcal{M}$ -valued functions on Ω ; in particular, we will often restrict to the degree-one subspace consisting of elements of the form

$$x = \sum_{i=1}^n \varepsilon_i \times x_i, \quad x_i \in \mathcal{M}.$$

For each i , let $\mathbb{E}_i : \mathcal{A} \rightarrow \mathcal{A}$ denote the conditional expectation onto the von Neumann subalgebra generated by $\{\varepsilon_j : j \neq i\}$, acting trivially on $\mathcal{M}$. We define the Lindbladian by

$$L = \sum_{i=1}^n L_i, \quad L_i = \text{Id} - \mathbb{E}_i.$$

One checks that the spectral gap of L is 1 and L is GNS- ψ -detailed balance, since each L_i is a projection; moreover $E_{fix} = \mathbb{E} \otimes \text{Id}_{\mathcal{M}}$ and $E_{fix}(x) = 0$. The associated gradient form for $x = \sum_{i=1}^n \varepsilon_i \times x_i$ decomposes as

$$\Gamma = \sum_{i=1}^n \Gamma_i, \quad \Gamma_i(x, x) = x_i^* x_i, \quad \Gamma_i(x^*, x^*) = x_i x_i^*.$$

Denote by h_ϕ the density of ϕ , and set

$$a = xh_\phi^{1/p} = \sum_{i=1}^n \varepsilon_i \times a_i, \quad a_i = x_i h_\phi^{1/p}.$$

Following Equation (23), $E_p^\phi(\cdot) = E(\cdot)h_\phi^{\frac{1}{p}}$ and thus $E_p^\phi(a) = 0$. By Theorem 4.18 and density, for all $a_i \in L^p(\mathcal{M})$ and $p = 2$ or $p \geq 3$,

$$\begin{aligned} \|a - E_p^\phi(a)\|_{L^p(\mathcal{A})} &= \left\| \sum_{i=1}^n \varepsilon_i \times a_i \right\|_{L^p(\mathcal{A})} \\ &\leq \frac{p}{\sqrt{2}} \left\| \left(\sum_{i=1}^n a_i^* a_i \right)^{1/2} \right\|_{L^p(\mathcal{M})} \\ &\quad + \frac{p}{\sqrt{2}} \left\| \left(\sum_{i=1}^n a_i a_i^* \right)^{1/2} \right\|_{L^p(\mathcal{M})}. \end{aligned}$$

Our application gives a suboptimal constant for noncommutative Khintchine inequalities in [35] with constant $O(\sqrt{p})$.

5.2 Sub-exponential concentration inequalities

It is well known that Poincaré inequalities imply concentration inequalities. We establish that our $\text{PI}(p, p)$ implies a sub-exponential concentration inequality, which in the semi-commutative case is established in [21, Thm 2.7].

To obtain the desired concentration inequality, we need to resolve the p -dependence of Γ_p , as a direct estimate is not immediate. We therefore return to the derivation framework of [27]. The following lemma is stated for a general tracial von Neumann algebra and will be applied below to each tracial slice $(\mathcal{M}_k, \tau_k)$.

We summarize the noncommutative derivation framework of [27, Prop. 2.6, Prop. 2.22] as below.

Lemma 5.1 (Noncommutative derivation). *Let $(\mathcal{M}, \tau)$ be a tracial von Neumann algebra and let*

$$L_\varepsilon = \frac{I - T_\varepsilon}{\varepsilon}$$

be the ε -regularized generator associated with a trace-symmetric QMS. There exists a free extension

$$\mathcal{M} \subset \mathcal{M} \otimes_{T_\varepsilon} \mathcal{M} \subset \mathcal{M}$$

associated with the Hilbert module construction [30] of T_ε , and a self-adjoint element $\xi_\varepsilon \in \mathcal{M}$, such that

$$\partial_\varepsilon(x) := [x, \xi_\varepsilon], \quad x \in \mathcal{M},$$

defines a derivation into $L^2(\mathcal{M})$. Then the following holds.

(i) For $x, y \in \mathcal{M}$,

$$2\Gamma_{L_\varepsilon}(x, y) = E_{\mathcal{M}}((\partial_\varepsilon(x))^* \partial_\varepsilon(y)).$$

(ii) For $2 \leq p \leq \infty$, there exist universal constants $C > 0$ such that for self-adjoint $x \in \mathcal{M}_{sa}$

$$\sqrt{2}\|\Gamma_{L_\varepsilon}(x, x)\|_{L^{p/2}(\mathcal{M})}^{1/2} \leq \|\partial_\varepsilon(x)\|_{L^p(\mathcal{M})} \leq C\|\Gamma_{L_\varepsilon}(x, x)\|_{L^{p/2}(\mathcal{M})}^{1/2}.$$

In particular,

$$\|\partial_\varepsilon(x)\|_\infty \leq 2\sqrt{2}\|\Gamma_{L_\varepsilon}(x, x)\|_\infty^{1/2}. \quad (32)$$

On the tracial slices $(\mathcal{M}_k, \tau_k)$, Theorem 4.22 gives for $\alpha \geq 0$, $T_{t,k}(d_k^\alpha x) = d_k^\alpha T_{t,k}(x)$, where $d_k \in \mathcal{M}_k$ is the density of $\phi_k := \widehat{\phi}|_{\mathcal{M}_k}$ with respect to τ_k . As $T_{t,k}$ is τ_k -symmetric, the L^p -version identifies with $T_{t,k}$ on self-adjoint elements

$$a = d_k^{1/2p} x d_k^{1/2p} \in L^p(\mathcal{M}_k), \quad x \in (\mathcal{M}_k)_{sa}.$$

The following corollary is also for a tracial von Neumann algebra $(\mathcal{M}, \tau)$, where $L^p(\mathcal{M})$ is identified with $\mathcal{M}$. Essentially, we apply the framework for $(\mathcal{M}_k, \tau_k)$ and drop the subscript k to be notationally clear.

Corollary 5.2. *Let $(\mathcal{M}, \tau)$ be finite and tracial, and let $d \in \mathcal{M}$ be the density of a normal faithful state ϕ with respect to τ . Assume that T_ε is τ -symmetric and*

$$T_\varepsilon(d^\alpha x) = d^\alpha T_\varepsilon(x), \quad \alpha \geq 0, \varepsilon > 0, x \in \mathcal{M}.$$

Following the notation of Lemma 5.1, one has

$$[\xi_\varepsilon, d^\alpha] = 0, \quad \alpha \geq 0.$$

Consequently, the derivation associated with the L^p -version of the Lindbladian $L_{\varepsilon,p}$ from Theorem 4.9 on dense elements is given by

$$\partial_\varepsilon(a) := [a, \xi_\varepsilon] = d^{1/2p} \partial_\varepsilon(x) d^{1/2p}, \quad a = d^{1/2p} x d^{1/2p}, \quad x \in \mathcal{M}_{sa}. \quad (33)$$

The L^p -gradient form eq. (24), by identification of $L_{\varepsilon,p}$ with L_ε on dense elements and item (i) of Lemma 5.1, satisfies

$$2\Gamma_{L_{\varepsilon,p}}(a, a) = E_{\mathcal{M}}(\partial_\varepsilon(a)^* \partial_\varepsilon(a)).$$

Proof. We use the free construction of $\mathcal{M}$ from [27, Section 2]. In this construction

$$\xi_\varepsilon = s(1_{[0,1/\varepsilon]}), \quad s(f) = \ell(f) + \ell(f)^*,$$

where $\ell(f)$ is the left creation operator used in the free construction of $\mathcal{M}$.

By the multiplicative-domain property of d^α ,

$$T_\varepsilon(d^\alpha x) = d^\alpha T_\varepsilon(x), \quad T_\varepsilon(x d^\alpha) = T_\varepsilon(x) d^\alpha.$$

Thus, in $\mathcal{M} \otimes_{T_\varepsilon} \mathcal{M}$, the left and right actions of d^α agree on the distinguished vector $\zeta = 1 \otimes 1$:

$$d^\alpha \zeta = \zeta d^\alpha.$$

Therefore, for every $f \in L^2(0, \infty)$,

$$d^\alpha(f \otimes \zeta) = (f \otimes \zeta) d^\alpha.$$

Since $\ell(f)$ is the left creation operator

$$\ell(f)(\zeta_1 \otimes \cdots \otimes \zeta_n) = (f \otimes \zeta) \otimes \zeta_1 \otimes \cdots \otimes \zeta_n,$$

the relation $d^\alpha(f \otimes \zeta) = (f \otimes \zeta) d^\alpha$ implies

$$[\ell(f), d^\alpha] = 0.$$

Taking the adjoint gives the same commutation relation for $\ell(f)^*$. Choosing $f = 1_{[0, 1/\varepsilon]}$, we get

$$[\xi_\varepsilon, d^\alpha] = 0.$$

Now let $a = d^{1/2p} x d^{1/2p}$. Since ξ_ε commutes with $d^{1/2p}$,

$$[a, \xi_\varepsilon] = d^{1/2p} [x, \xi_\varepsilon] d^{1/2p} = d^{1/2p} \partial_\varepsilon(x) d^{1/2p}.$$

This gives the formula for $\partial_\varepsilon(a)$. The identity for $\Gamma_{L_{\varepsilon,p}}$ follows from item (i) of Lemma 5.1 applied to the L^p -gradient form eq. (24):

$$2\Gamma_{L_{\varepsilon,p}}(a, a) = E_{\mathcal{M}}([a, \xi_\varepsilon]^* [a, \xi_\varepsilon]) = E_{\mathcal{M}}(\partial_\varepsilon(a)^* \partial_\varepsilon(a)).$$

□

We can establish the following Corollary for $\text{PI}(p, \infty)$.

Corollary 5.3 ($\text{PI}(p, \infty)$ with $O(p)$ constant). *Given the same conditions as the Theorem 4.18, for $a := h_\phi^{\frac{n}{p}} x h_\phi^{\frac{1-\eta}{p}} \in L^p(\mathcal{M})$, $x \in \mathcal{M}_a^\sigma$,*

$$\|a - E_p^\phi(a)\|_{L^p(\mathcal{M})} \leq 2p \sqrt{\frac{2}{\alpha}} \|x\|_{\text{Lip}_\Gamma},$$

where the Lipschitz seminorm [15] is $\|x\|_{\text{Lip}_\Gamma} := \max\{\|\Gamma(x, x)\|_\infty^{1/2}, \|\Gamma(x^*, x^*)\|_\infty^{1/2}\}$.

Proof. To use the derivation trick, we first consider $\eta = \frac{1}{2}$ and self-adjoint $x = x^*$. The case of general $\eta \in [0, 1]$ and $x \in \mathcal{M}_a^\sigma$ follows by similar decomposition into self-adjoint and skew-adjoint parts as in Theorem 3.9, with an extra factor 2. On the slice $(\mathcal{M}_k, \tau_k)$, let

$$L_\varepsilon^k := \frac{I - T_\varepsilon^k}{\varepsilon}$$

be the ε -regularized generator. Its spectral gap is $\alpha_\varepsilon = \frac{1-e^{-\alpha\varepsilon}}{\varepsilon}$ by Theorem C.1. The PI(p, p) inequality takes the form

$$\|a_k - E_k(a_k)\|_{L^p(\mathcal{M}_k)} \leq \frac{p}{\sqrt{2\alpha_\varepsilon}} \|a_k\|_{\Gamma_{L_\varepsilon^k, p}^k}.$$

The right-hand side can be expressed in terms of the derivation ∂_ε^k constructed in Theorem 5.2. In Haagerup's L^p notation, writing $\Gamma_p^k = \Gamma_{L_\varepsilon^k, p}^k$, we have

$$\begin{aligned} \sqrt{2}\|a_k\|_{\Gamma_p^k} &= \left\| E_{\mathcal{M}_k}((\partial_\varepsilon^k(a_k))^* \partial_\varepsilon^k(a_k)) \right\|_{L^{p/2}(\mathcal{M}_k)}^{1/2} \\ &\leq \|\partial_\varepsilon^k(a_k)\|_{L^p(\mathcal{M}_k)} \\ &= \|d_k^{1/2p} \partial_\varepsilon^k(x_k) d_k^{1/2p}\|_{L^p(\mathcal{M}_k)} \\ &\leq \|d_k^{1/2p}\|_{L^{2p}(\mathcal{M}_k)}^2 \|\partial_\varepsilon^k(x_k)\|_\infty = \|\partial_\varepsilon^k(x_k)\|_\infty \\ &\leq 2\sqrt{2}\|\Gamma_{L_\varepsilon}^k(x_k, x_k)\|_\infty^{1/2}. \end{aligned}$$

The first equality follows from Theorem 5.2. The next inequality uses the contractivity of conditional expectations. The third line uses eq. (33). The fourth line follows from Hölder's inequality and $\tau_k(d_k) = 1$ since d_k is density of a normal faithful state ϕ_k . The last inequality follows from eq. (32).

Since the estimate is uniform in $k \geq 1$, we argue as in the proof of Theorem 4.17 and take $\varepsilon \downarrow 0$. Passing to the Haagerup reduction limit $k \rightarrow \infty$ yields the desired estimate. $\square$

Remark 5.4. Corollary 5.3 may be compared with [16, Theorem 5.12], where a PI(p, ∞) inequality with constant $O(\sqrt{p})$ is obtained under the assumption $\alpha_{\text{MLSI}} > 0$, rather than an L^2 spectral gap. In view of the discussion in the introduction, the two results should be regarded as independent.

We follow the procedure of [16] for the sub-exponential concentration property.

Definition 5.5. For $x \in \mathcal{M}$, we write

$$\mathbb{P}_\phi(|x| > t) \leq \epsilon$$

if there exists a projection $e \in \mathcal{M}$ such that

$$\|exe\|_\infty \leq t \quad \text{and} \quad \phi(1 - e) \leq \epsilon.$$

Lemma 5.6 ([16, Lemma 5.14]). Let $x \in L_p(\mathcal{M}, \phi)$ with $1 < p < \infty$. Then

$$\mathbb{P}_\phi(|x| > t) \leq 2 \left(\frac{t}{4}\right)^{-p} \|x\|_{L^p(\mathcal{M})}^p.$$

Corollary 5.7 (Sub-exponential Concentration). *Let $T_t = e^{-tL}$ be a GNS- ϕ -detailed balance QMS with spectral gap $\alpha > 0$ and ϕ -preserving conditional expectation E^ϕ onto its fixed-point algebra. Then for any $x \in \mathcal{M}$ and large $t > 0$,*

$$\mathbb{P}_\phi(|x - E^\phi(x)| > t) \leq 2 \exp\left(-\frac{\sqrt{\alpha}t}{8\sqrt{2}e\|x\|_{\text{Lip}_\Gamma}}\right).$$

Proof. By Lemma 5.6 and Corollary 5.3,

$$\mathbb{P}_\phi(|x - E(x)| > t) \leq 2\left(\frac{t}{4}\right)^{-p} \|x - E(x)\|_{L_p(\mathcal{M})}^p \leq 2\left(\frac{8\sqrt{2}p\|x\|_{\text{Lip}_\Gamma}}{\sqrt{\alpha}t}\right)^p.$$

Optimizing in p gives $p = \frac{\sqrt{\alpha}t}{8\sqrt{2}e\|x\|_{\text{Lip}_\Gamma}} \geq 3$ for t large, which implies the claim. $\square$

This is a generalization of the sub-exponential concentration obtained in the semi-commutative case [21, Thm 2.7]. Under only spectral gaps, one cannot expect tail decay faster than exponential, since this hypothesis does not imply stronger concentration phenomena. In contrast, [26, Cor 5.15] proved Gaussian concentration under the additional assumption of MLSI, which is strictly stronger.

5.2.1 Birth–death process

The following birth-death process shows that the $O(p)$ PI(p, p) inequality in Theorem 1.3 is weaker than the $O(\sqrt{p})$ Lipschitz-moment estimate under an MLSI assumption [16, Theorem 5.12]. We illustrate this by a violation of the geometric Talagrand inequality by considering a birth-death process with a constant spectral gap. However, by [15, Thm 6.10 and Rem. 6.9], $O(\sqrt{p})$ Lipschitz-moment estimate implies a Wasserstein-1 bound and hence geometric Talagrand concentration. Therefore, the $O(\sqrt{p})$ estimate cannot follow from a spectral gap alone, whereas the $O(p)$ estimate in Theorem 1.3 still holds.

Definition 5.8 (Geometric Talagrand inequality). *Let $T_t = e^{-tL}$ be a QMS on $(\mathcal{M}, \tau)$ with fixed-point algebra $\mathcal{N}$ and conditional expectation $E: \mathcal{M} \rightarrow \mathcal{N}$. We say that T_t satisfies the geometric Talagrand inequality if there exists $c > 0$ such that for every normal state ρ ,*

$$\|\rho - E_*(\rho)\|_{\Gamma^*} \leq c\sqrt{D(\rho\|E_*(\rho))}, \quad (34)$$

where $\|\rho\|_{\Gamma^*} := \sup\{|\tau(\rho x)| : E(x) = 0, x = x^*, \|x\|_{\text{Lip}_\Gamma} \leq 1\}$ and E_* is the predual of E .

Example 5.9 (Birth-death Process). *For a 1D chain of length n and fixed $\beta > 0$, we define the birth-death process edge-wise as follows*

$$\begin{aligned} L_{j,j+1}(x) = & e^{\beta/2}(|j+1\rangle\langle j+1|x + x|j+1\rangle\langle j+1| - 2|j+1\rangle\langle j|x|j\rangle\langle j+1|) \\ & + e^{-\beta/2}(|j\rangle\langle j|x + x|j\rangle\langle j| - 2|j\rangle\langle j+1|x|j+1\rangle\langle j|). \end{aligned}$$

The total birth-death Lindbladian is

$$L = \sum_{j=1}^{n-1} L_{j,j+1}.$$

The Lindbladian L satisfies GNS- μ detailed balance for the thermal state $\mu := Z_\beta^{-1} \sum_{k=1}^n e^{-\beta k} |k\rangle \langle k|$, where $Z_\beta := \sum_{j=1}^n e^{-\beta j}$ is the partition function. Restricting to the diagonal part L^{diag} and fixed $\beta > 0$, [37] gives the spectral gap $\alpha(L^{\text{diag}}) = O(1)$. We therefore establish $\text{PI}(p, p)$ with constant $O(p)$ by Theorem 1.3.

Nevertheless, the Talagrand's inequality Equation (34) admits no constant $c = c(\beta)$ independent of n . Following the setup in [15, Remark 6.9], consider e_A a projection in region A and density matrix $\rho_A := \frac{e_A}{\text{Tr}(e_A)}$. Choose $f = \sum_{j=1}^n \frac{j}{\sqrt{2 \cosh \beta/2}} |j\rangle \langle j|$ which gives

$$\|f\|_{\text{Lip}_\Gamma} = \left\| \sum_j \left[e^{\beta/4} (|j\rangle \langle j+1|), f \right]^2 + \sum_j \left[e^{-\beta/4} (|j+1\rangle \langle j|), f \right]^2 \right\|^{1/2} = 1.$$

Let $A = \{1\}$ and $B = \{n\}$. Since ρ_A and ρ_B are supported on disjoint A and B ,

$$\|\rho_A - \rho_B\|_{\Gamma^*} = \sup_{x=x^*, \|x\|_{\text{Lip}_\Gamma} \leq 1} |\text{Tr}((\rho_A - \rho_B)x)| \geq |\text{Tr}((\rho_A - \rho_B)f)| = \frac{n-1}{\sqrt{2 \cosh \beta/2}}.$$

Moreover,

$$D(\rho_{\{i\}} \| E_*(\rho_{\{i\}})) = D(\rho_{\{i\}} \| \mu) = -\ln \mu(\{i\}), \quad i \in \{1, \dots, n\}.$$

The right-hand side $c(\beta)(\sqrt{D(\rho_A \| E_*(\rho_A))} + \sqrt{D(\rho_B \| E_*(\rho_B))}) = O(\sqrt{n})$, whereas the left-hand side $\|\rho_A - \rho_B\|_{\Gamma^*}$ of linear order in n . This contradicts Equation (34) for any fixed $c = c(\beta)$ as $n \rightarrow \infty$.

The birth-death process above provides an example showing that a subgaussian-type estimate, where K is a universal constant

$$\|f - E(f)\|_p \leq K c(\beta)^2 \sqrt{p} \|f\|_{\text{Lip}_\Gamma},$$

fails uniformly in n with a constant depending only on β .

This agrees with the result [16, Thm 5.7] where they established the CMLSI $\alpha_c = O(1/n)$ replacing $c(\beta)^2$. In contrast, L^{diag} has a spectral gap of $O(1)$, so the $\text{PI}(p, p)$ inequality with constant $O(p)$ still holds.

5.3 Lipschitz diameter

As an application of Theorem 1.3, we obtain Lipschitz diameter and completely bounded (CB) Lipschitz diameter estimates for GNS- ϕ -detailed-balanced QMSs in finite dimensions, depending on the spectral gap α and the spectral density of ϕ .

Corollary 5.10 (Lipschitz diameter). *Let $\mathbb{M}_d$ be a finite-dimensional matrix algebra equipped with the unnormalised trace Tr and large dimension (e.g. $d > 20$), and denote by $\|\cdot\|_{S_p} := (\text{Tr}|\cdot|^p)^{1/p}$ the Schatten- p norm. Let $(T_t)_{t \geq 0}$ be a quantum Markov semigroup satisfying GNS detailed balance with respect to $\phi(\cdot) = \text{Tr}(D_\phi \cdot)$ and spectral gap $\alpha > 0$. Then for any $x \in \mathbb{M}_d$,*

$$\|x - E(x)\|_\infty \leq \text{diam}_\Gamma \|x\|_{\text{Lip}_\Gamma},$$

where $\text{diam}_\Gamma \leq 2e \log(\lambda_{\min}^{-1}) \sqrt{\frac{2}{\alpha}}$ and $\lambda_{\min}$ denotes the minimal eigenvalue of D_ϕ .

Moreover, invoking Pisier's vector-valued Schatten norm [41] at each matrix level, the corresponding completely bounded diameter estimate is

$$\text{diam}_\Gamma^{\text{cb}} := \sup_{m \geq 1} \text{diam}_{\Gamma^{(m)}} \leq 4e \log(\lambda_{\min}^{-1}) \sqrt{\frac{2}{\alpha}}.$$

Proof. We write D as shorthand for D_ϕ and, without loss of generality, $x \in \mathring{\mathbb{M}}_d$, where $\mathring{\mathbb{M}}_d = \ker(E)$. Since $x = D^{-\frac{1}{2p}}(D^{\frac{1}{2p}} x D^{\frac{1}{2p}}) D^{-\frac{1}{2p}}$, the Hölder inequality and Corollary 5.3 gives

$$\begin{aligned} \|x\|_\infty &\leq \|x\|_{S_p} \leq \|D^{-\frac{1}{2p}}\|_\infty^2 \|D^{\frac{1}{2p}} x D^{\frac{1}{2p}}\|_{S_p} \\ &\leq (\lambda_{\min})^{-\frac{1}{p}} \|\iota_{1/2}(x)\|_{p, \phi, 1/2} \leq (\lambda_{\min})^{-\frac{1}{p}} 2p \sqrt{\frac{2}{\alpha}} \|x\|_{\text{Lip}_\Gamma}, \end{aligned}$$

Now choose $p = \log(\lambda_{\min}^{-1}) \geq \log(d) > 3$, so that $(\lambda_{\min})^{-1/p} = e$. Substituting back leads to the desired inequality.

For the complete version, fix $m \geq 1$ and consider the amplified semigroup $\text{Id}_m \otimes T_t$. Its fixed-point algebra is $\mathbb{M}_m \overline{\otimes} \mathcal{N}$, and its conditional expectation onto this algebra is $\text{Id}_m \otimes E$. Let $\Gamma^{(m)}$ denote the gradient form of $\text{Id}_m \otimes T_t$. Let $X \in \mathbb{M}_m(\mathbb{M}_d)$ be self-adjoint and satisfy

$$X = X^* \in \ker(\text{Id}_m \otimes E) \subset \mathbb{M}_m(\mathbb{M}_d).$$

For every $p \geq 3$, Pisier's formula for self-adjoint elements [41] gives

$$\|X\|_{\mathbb{M}_m(\mathbb{M}_d)} = \sup_{\psi} \|(D_\psi^{1/(2p)} \otimes 1) X (D_\psi^{1/(2p)} \otimes 1)\|_{S_p^m[\mathbb{M}_d]}, \quad (35)$$

where the supremum runs over the states ψ on $\mathbb{M}_m$ and D_ψ denotes the density of ψ with respect to Tr_m . i.e. $\psi(\cdot) = \text{Tr}_m(D_\psi \cdot)$.

To estimate the RHS of eq. (35), we can expand $1 = D^{\frac{1}{2p}} D^{-\frac{1}{2p}}$, and rewrite

$$(D_\psi^{1/(2p)} \otimes 1)X(D_\psi^{1/(2p)} \otimes 1) = (\text{Id}_{S_p^m} \otimes \alpha_p)(Z_\psi),$$

where $Z_\psi := (D_\psi^{1/(2p)} \otimes D^{1/(2p)})X(D_\psi^{1/(2p)} \otimes D^{1/(2p)})$ and the map α_p is defined as

$$\alpha_p : S_p^d \longrightarrow \mathbb{M}_d, \quad \alpha_p(Y) = D^{-1/(2p)} Y D^{-1/(2p)}.$$

The completely bounded two-sided multiplication estimate for the canonical operator-space structure on S_p^d [42] yields

$$\|\alpha_p\|_{\text{cb}(S_p^d, \mathbb{M}_d)} \leq \|D^{-1/(2p)}\|_{S_{2p}}^2 = \left(\sum_j \lambda_j^{-1}\right)^{1/p}. \quad (36)$$

Using eq. (36) and the complete isometry $S_p^m[S_p^d] = S_p^{md}$, we obtain estimate of RHS of eq. (35):

$$\|(D_\psi^{1/(2p)} \otimes 1)X(D_\psi^{1/(2p)} \otimes 1)\|_{S_p^m[\mathbb{M}_d]} \leq \left(\sum_j \lambda_j^{-1}\right)^{1/p} \|Z_\psi\|_{S_p^{md}}. \quad (37)$$

The amplified semigroup $\text{Id}_m \otimes T_t$ has spectral gap α relative to $\text{Id}_m \otimes E$ by Theorem B.1. Since $X = X^*$ and $(\text{Id}_m \otimes E)(X) = 0$, the self-adjoint $\eta = \frac{1}{2}$ estimate from the proof of Corollary 5.3, applied to the state $\psi \otimes \phi$, gives

$$\|Z_\psi\|_{S_p^{md}} \leq p \sqrt{\frac{2}{\alpha}} \|X\|_{\text{Lip}_{\Gamma(m)}}.$$

Combining this estimate with eqs. (35) and (37) and then taking the supremum over ψ yields

$$\|X\|_{\mathbb{M}_m(\mathbb{M}_d)} \leq p \left(\sum_j \lambda_j^{-1}\right)^{1/p} \sqrt{\frac{2}{\alpha}} \|X\|_{\text{Lip}_{\Gamma(m)}}. \quad (38)$$

Set $A = \sum_j \lambda_j^{-1}$ and choose $p = \log A$ to minimize the RHS of eq. (38). We have

$$\text{diam}_\Gamma^{\text{cb}} \leq e \log A \sqrt{\frac{2}{\alpha}} \leq 2e \log(\lambda_{\min}^{-1}) \sqrt{\frac{2}{\alpha}},$$

where we used $A \leq d\lambda_{\min}^{-1} \leq \lambda_{\min}^{-2}$ in the last inequality. Finally, for a general X , the decomposition into its self-adjoint and skew-adjoint parts, as in Theorem 3.9, contributes a factor 2. Taking the supremum over $m \geq 1$ therefore yields the desired completely bounded estimate. $\square$

Returning to the birth-death process in Example 5.9 for fixed $\beta > 0$, we can apply the above corollary to obtain a diameter estimate for the semigroup. The minimal eigenvalue of the thermal state μ is $\lambda_{\min} = Z_{\beta}^{-1}e^{-\beta n}$, which gives

$$\text{diam}_{\Gamma} = O(n), \quad \text{diam}_{\Gamma}^{\text{cb}} = O(n).$$

A Markov dilation and proof of Theorem 3.7

Before proving the noncommutative convex-chain estimate for gradient forms, we first recall the framework of Markov dilation, which lifts a quantum Markov semigroup to a multiplicative level. This allows the Dirichlet form to be written as a square and enables the noncommutative chain rule.

Markov dilation embeds quantum Markov semigroups into larger operator-algebraic structures that carry probabilistic interpretations. The foundational work of Kümmerer and Maassen [29] introduced noncommutative dilation in which the enlarged algebra remains essentially commutative, and the first author and Mei [24] extended this framework to the setting of tracial von Neumann algebras. In [52], the author provided a probabilistic construction of Markov dilation. In quantum physics, such dilations are interpreted as providing a reversible microscopic model of systems interacting with thermal environments. We recall the definition from [24, Section 2.2].

Definition A.1 (Markov dilation). *Let $(T_t)_{t \geq 0}$ be a quantum Markov semigroup on a tracial von Neumann algebra $(\mathcal{M}, \tau)$. We say that (T_t) admits a Markov dilation if there exists an increasing family of tracial von Neumann algebras*

$$\mathcal{M} \subset \widetilde{\mathcal{M}}_t \subset \widetilde{\mathcal{M}}, \quad t \geq 0,$$

that is, an increasing filtration of $\widetilde{\mathcal{M}}_t$, together with

- *normal, trace-preserving conditional expectations $E_t : \widetilde{\mathcal{M}} \rightarrow \widetilde{\mathcal{M}}_t$,*
- *$*$ -homomorphisms $\pi_t : \mathcal{M} \rightarrow \widetilde{\mathcal{M}}$ that are adapted to the filtration, i.e. $\pi_t(\mathcal{M}) \subset \widetilde{\mathcal{M}}_t$, for each $t \geq 0$,*

such that for all $0 \leq s < t < \infty$ and all $x \in \mathcal{M}$,

$$E_s(\pi_t(x)) = \pi_s(T_{t-s}(x)). \tag{39}$$

Equivalently, condition (39) is expressed by the following commuting diagram.

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{\pi_t} & \widetilde{\mathcal{M}} \\ T_{t-s} \downarrow & & \downarrow E_s \\ \mathcal{M} & \xrightarrow{\pi_s} & \widetilde{\mathcal{M}}_s \end{array} \tag{40}$$

We now prove Theorem 3.7, restated below for convenience.

Proposition (Convex-chain estimate for gradient form). *Given the conditions of Section 2 and lemma 3.6, for every self-adjoint element $x \in \text{Dom}(L) \subset L_2(\mathcal{M}, \tau)$,*

$$\mathcal{E}(\varphi(x)) \leq \tau(\Gamma(x, x)\psi(x)),$$

where $\mathcal{E}(f) := \tau(\Gamma(f, f))$ is the Dirichlet form associated with L .

Proof. Use Markov dilation from Theorem A.1, for $x \in \mathcal{M}$, and from the commuting diagram (40) with $s = 0$ $T_t = E_0 \circ \pi_t|_{\mathcal{M}}$ and that T_t is τ -symmetric as in equation (6),

$$\begin{aligned} \mathcal{E}(x) &:= \tau(x^* L(x)) = \lim_{t \rightarrow 0} \tau\left(x^* \frac{1 - e^{-tL}}{t} x\right) = \lim_{t \rightarrow 0} \frac{\tau(x^* x) - \tau(x^* T_t(x))}{t} \\ &= \lim_{t \rightarrow 0} \frac{1}{2t} \tau\left(x^* x + T_t(x^* x) - T_t(x)^* x - x^* T_t(x)\right) \\ &= \lim_{t \rightarrow 0} \frac{1}{2t} \tau\left(x^* x + \pi_t(x^* x) - \pi_t(x)^* x - x^* \pi_t(x)\right) \\ &= \lim_{t \rightarrow 0} \frac{1}{2t} \tau\left((x - \pi_t(x))^* (x - \pi_t(x))\right). \end{aligned}$$

Here the first line is the usual definition of the generator equation (5). The second line uses the τ -symmetry of the semigroup, namely $\tau(x^* T_t(x)) = \tau(T_t(x)^* x)$ and $\tau(T_t(x^* x)) = \tau(T_t(1)^* x^* x) = \tau(x^* x)$. In the third line we replace T_t by $E_0 \pi_t$ and drop E_0 inside the trace because $\tau \circ E_0 = \tau$ and $x \in \mathcal{M} = \widetilde{\mathcal{M}}_0$ so $E_0(x) = x$. Finally, since π_t is a $*$ homomorphism, we reorganize the expression into the squared difference in the last line. Now restrict to $x \in \mathcal{M}_{sa}$, so that $\varphi(x), \psi(x) \in \mathcal{M}_{sa}$. We then apply the

same argument to $\varphi(x)$ and use Lemma 3.6 to obtain

$$\begin{aligned}
\mathcal{E}(\varphi(x)) &= \lim_{t \rightarrow 0} \frac{1}{2t} \tau \left((\varphi(x) - \pi_t(\varphi(x)))^2 \right) \\
&= \lim_{t \rightarrow 0} \frac{1}{2t} \tau \left((\varphi(x) - \varphi(\pi_t(x)))^2 \right) \\
&\leq \lim_{t \rightarrow 0} \frac{1}{4t} \tau \left((x - \pi_t(x))^2 (\psi(x) + \psi(\pi_t(x))) \right) \\
&= \lim_{t \rightarrow 0} \frac{1}{4t} \tau \left(x^2 \psi(x) - x \pi_t(x) \psi(x) - \pi_t(x) x \psi(x) + \pi_t(x)^2 \psi(x) \right) \\
&\quad + \lim_{t \rightarrow 0} \frac{1}{4t} \tau \left(x^2 \pi_t(\psi(x)) - x \pi_t(x) \pi_t(\psi(x)) - \pi_t(x) x \pi_t(\psi(x)) + \pi_t(x)^2 \pi_t(\psi(x)) \right) \\
&= \lim_{t \rightarrow 0} \frac{1}{4t} \tau \left(x^2 \psi(x) - T_t(x) \psi(x) x - T_t(x) x \psi(x) + T_t(x^2) \psi(x) \right) \\
&\quad + \lim_{t \rightarrow 0} \frac{1}{4t} \tau \left(x^2 \pi_t(\psi(x)) - x \pi_t(x \psi(x)) - x \pi_t(\psi(x) x) + \pi_t(x^2 \psi(x)) \right) \\
&= \lim_{t \rightarrow 0} \frac{1}{4t} \tau \left((x^2 - x T_t(x) - T_t(x) x + T_t(x^2)) \psi(x) \right) \\
&\quad + \lim_{t \rightarrow 0} \frac{1}{4t} \tau \left(T_t(x^2) \psi(x) - T_t(x) x \psi(x) - x T_t(x) \psi(x) + x^2 \psi(x) \right) \\
&= \tau(\Gamma(x, x) \psi(x)),
\end{aligned}$$

In these manipulations we repeatedly use the cyclicity of the trace, the multiplicativity and trace preservation of π_t , the τ -symmetry of T_t , and $\tau \circ E_0 = \tau$. After canceling the mixed terms with those properties, the remaining t -difference quotient is recognized as the gradient form $\Gamma(x, x)$ defined in (7). Notice that we *never* assumed $\pi_t(x)$ and x to be exchangeable; π_t is not required to be an automorphism on $\mathcal{M}$. The argument relies only on the elementary properties already listed (multiplicativity of π_t and trace preservation), and not on any further commutation relations. The price for working under these minimal hypotheses is the lengthy algebraic expansion needed to isolate the gradient form term. $\square$

B Spectral gaps in AFP

This section identifies the spectral gap for tensor, direct-sum, and amalgamated product quantum Markov semigroups.

Proposition B.1 (Spectral gap for $T_t^1 \otimes T_t^2$). *Let $(T_t^i)_{t \geq 0}$ be quantum Markov semigroups on $\mathcal{M}_i$ with conditional expectations $E_i : \mathcal{M}_i \rightarrow \mathcal{N}_i$ onto the fixed-point algebras $\mathcal{N}_i$ with spectral gaps $\alpha_i > 0$, i.e.*

$$\|T_t^i(x) - E_i(x)\|_2 \leq e^{-\alpha_i t} \|x - E_i(x)\|_2, \quad x \in \mathcal{M}_i, t \geq 0.$$

On $\mathcal{M} := \mathcal{M}_1 \otimes \mathcal{M}_2$ consider the product semigroup $T_t := T_t^1 \otimes T_t^2$, with conditional expectation $E := E_1 \otimes E_2 : \mathcal{M}_1 \otimes \mathcal{M}_2 \rightarrow \mathcal{N}_1 \otimes \mathcal{N}_2$. Then

$$\|T_t(x) - E(x)\|_2 \leq e^{-\min\{\alpha_1, \alpha_2\}t} \|x - E(x)\|_2, \quad x \in \mathcal{M}, t \geq 0,$$

i.e. the spectral gap of $T_t^1 \otimes T_t^2$ is $\min\{\alpha_1, \alpha_2\}$.

Proof. Regard E_i as the orthogonal projection from $\mathcal{H}_i := L^2(\mathcal{M}_i, \tau_i)$ onto $L^2(\mathcal{N}_i, \tau_i)$, and set $\mathring{\mathcal{H}}_i := \ker E_i$. Since $T_t^i E_i = E_i T_t^i = E_i$, the orthogonal decomposition

$$\ker(E_1 \otimes E_2) = (\mathring{\mathcal{H}}_1 \otimes \mathcal{H}_2) \oplus (\text{Ran } E_1 \otimes \mathring{\mathcal{H}}_2)$$

reduces $T_t - E$. On its two summands, the restrictions of $T_t - E$ are, respectively,

$$(T_t^1 - E_1) \otimes T_t^2 \quad \text{and} \quad E_1 \otimes (T_t^2 - E_2).$$

It follows from the L^2 -contractivity of T_t^1 and T_t^2 that

$$\begin{aligned} \|T_t - E\|_{B(\ker E)} &\leq \max\{\|T_t^1 - E_1\|, \|T_t^2 - E_2\|\} \\ &\leq e^{-\min\{\alpha_1, \alpha_2\}t}. \end{aligned}$$

This proves the asserted estimate.

Finally, the restrictions of $T_t - E$ to $\mathring{\mathcal{H}}_1 \otimes \text{Ran } E_2$ and $\text{Ran } E_1 \otimes \mathring{\mathcal{H}}_2$ coincide with $T_t^1 - E_1$ and $T_t^2 - E_2$, respectively. Hence the spectral gap of the product semigroup cannot exceed either α_1 or α_2 . The preceding estimate therefore shows that it is exactly $\min\{\alpha_1, \alpha_2\}$. $\square$

Proposition B.2 (Spectral gap in $\mathcal{M}_1 \oplus \mathcal{M}_2$ via semigroups). *Under the setup of Theorem B.1, on $\mathcal{M} := \mathcal{M}_1 \oplus \mathcal{M}_2$ define $T_t := T_t^1 \oplus T_t^2$ and $E := E_1 \oplus E_2$. Then*

$$\|T_t(x) - E(x)\|_2 \leq e^{-\min\{\alpha_1, \alpha_2\}t} \|x - E(x)\|_2 \quad x \in \mathcal{M}_i, t \geq 0,$$

i.e. the spectral gap is $\min\{\alpha_1, \alpha_2\}$.

Proof. For $x = (x_1, x_2)$ in $\mathcal{H}$, $T_t(x) - E(x) = (T_t^1(x_1) - E_1(x_1), T_t^2(x_2) - E_2(x_2))$, so we have

$$\begin{aligned} \|T_t(x) - E(x)\|_2^2 &= \sum_{i=1}^2 \|T_t^i(x_i) - E_i(x_i)\|_2^2 \\ &\leq \sum_{i=1}^2 e^{-2\alpha_i t} \|x_i - E_i(x_i)\|_2^2 \leq e^{-2\min\{\alpha_1, \alpha_2\}t} \|x - E(x)\|_2^2. \end{aligned}$$

Taking square roots gives the desired estimate. $\square$

Corollary B.3 (Spectral gap in $\mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$ via semigroups). *Under the setup of Theorem B.1, let $\mathcal{M} := \mathcal{M}_1 *_{\mathcal{N}} \mathcal{M}_2$ carry the free-product semigroup T_t [8] with joint expectation E . Then for all $x \in L_2(\mathcal{M})$,*

$$\|T_t(x) - E(x)\|_2 \leq e^{-\min\{\alpha_1, \alpha_2\}t} \|x - E(x)\|_2, \quad x \in \mathcal{M}, t \geq 0$$

and hence the spectral gap is $\min\{\alpha_1, \alpha_2\}$.

Proof. Recall the orthogonal decomposition of the GNS-space (15)

$$L_2(\mathcal{M}) = L_2(\mathcal{N}) \oplus \bigoplus_{\substack{w=i_1 \dots i_k \\ i_j \in \{1,2\}, i_j \neq i_{j+1}}} \mathcal{H}_w,$$

where each $\mathcal{H}_w$ is the closed linear span of reduced words $x_{i_1} \dots x_{i_k}$ with $E_{i_j}(x_{i_j}) = 0$. On each summand $\mathcal{H}_w$, T_t acts by

$$T_t(x_{i_1} \dots x_{i_k}) = \pi_{i_1} \circ T_t^{(i_1)}(x_{i_1}) \dots \pi_{i_k} \circ T_t^{(i_k)}(x_{i_k}),$$

For simplicity, we give the proof in the scalar free-product case $\mathcal{N} = \mathbb{C}1$. For general $\mathcal{N}$, the same proof applies to the AFP decomposition by replacing ordinary tensor products with relative tensor products $\bar{\otimes}_{\mathcal{N}}$ and using $\mathcal{N}$ -bimodularity together with the induced tensor-operator norm estimate.

By orthogonality and the tensor-product estimate (Theorem B.1),

$$\|T_t(x_{i_1} \dots x_{i_k})\|_2 = \prod_{j=1}^k \|T_t^{(i_j)}(x_{i_j})\|_2 \leq \prod_{j=1}^k (e^{-\alpha_{i_j}t} \|x_{i_j}\|_2) = e^{-(\sum_j \alpha_{i_j})t} \|x_{i_1} \dots x_{i_k}\|_2.$$

In particular, the slowest decay on a word arises from its last letter, so

$$\|T_t|_{\mathcal{H}_w}\| \leq e^{-\min\{\alpha_{i_k}\}t} \leq e^{-\min\{\alpha_1, \alpha_2\}t}.$$

Since $L_2(\mathring{\mathcal{M}}) = L_2(\mathcal{M}) \ominus L_2(\mathcal{N})$ is the orthogonal direct sum of all the $\mathcal{H}_w$, we conclude

$$\|T_t - E\|_{B(L_2(\mathcal{M}) \ominus L_2(\mathcal{N}))} = \sup_w \|T_t|_{\mathcal{H}_w}\| \leq e^{-\min\{\alpha_1, \alpha_2\}t}.$$

Equivalently,

$$\|T_t(x) - E(x)\|_2 \leq e^{-\min\{\alpha_1, \alpha_2\}t} \|x - E(x)\|_2,$$

and by definition this decay rate and Prop. B.2 is exactly the spectral gap $\min\{\alpha_1, \alpha_2\}$. $\square$

C Properties of ϵ -regularized Lindbladians

C.1 Spectral gaps

Under the notation of the main text, let L be the generator of the semigroup $(T_t)_{t \geq 0}$, satisfying detailed balance (either GNS or with respect to the trace), and let $\alpha > 0$ denote its spectral gap. We verify that if $x \in \text{Dom}(L)$, that the ϵ -regularization

$$L_\epsilon(x) := \frac{L}{1 + \epsilon L}(x) \xrightarrow{\epsilon \downarrow 0} L(x)$$

in the strong* operator topology. We also verify that L_ϵ inherit the detailed balance property of L and satisfy the lower bound $\alpha_\epsilon \geq \alpha(1 - \epsilon\alpha)$ for their spectral gaps.

The preservation of detailed balance follows directly from the integral representation [11, Prop. 2.5],

$$L_\epsilon = \frac{1}{\epsilon}(I - R_\epsilon) = \frac{1}{\epsilon}\left(I - \int_0^\infty e^{-t} T_{\epsilon t} dt\right),$$

since $(T_t)_{t \geq 0}$ satisfy detailed balance.

For the spectral gap, we restrict to $(\mathcal{M}_k, \tau_k)$ with $k \geq 1$, where the $\text{PI}(p, p)$ inequality holds. Since $L_{k, \epsilon}$ is τ_k -symmetric, we apply spectral theory. If $\text{Spec}(L_k|_{\text{Ran } L_k}) \subset [\alpha, \infty)$, then by functional calculus

$$\text{Spec}(L_{k, \epsilon}|_{\text{Ran } L_{k, \epsilon}}) = \left\{ \frac{\lambda}{1 + \epsilon\lambda} : \lambda \in \text{Spec}(L_k) \right\} \subset \left[\frac{\alpha}{1 + \epsilon\alpha}, \infty \right).$$

Hence, the spectral gap gives

$$\alpha_\epsilon = \frac{\alpha}{1 + \epsilon\alpha} \geq \alpha(1 - \epsilon\alpha),$$

where the last inequality follows from $\frac{1}{1+x} \geq 1 - x$ for $x \geq 0$.

Remark C.1. We can also repeat the spectral gaps for the ε -regularization in [27], where

$$L_\varepsilon(x) := \frac{1 - T_\varepsilon(x)}{\varepsilon} \xrightarrow{\varepsilon \downarrow 0} L(x),$$

in the strong* operator topology for $x \in \text{Dom}(L)$. Assuming L have spectral gap $\alpha > 0$, the corresponding spectral gap gives

$$\alpha_\varepsilon = \frac{1 - e^{-\alpha\varepsilon}}{\varepsilon} \geq \frac{\alpha}{1 + \varepsilon\alpha},$$

which converges to α as $\varepsilon \downarrow 0$.

Moreover, for a convex family $\bar{L} = \sum_j \lambda_j L_{\epsilon_j}$ with $\epsilon_j > 0$, $\lambda_j \geq 0$, and $\sum_j \lambda_j = 1$, detailed balance holds since convex combinations of detailed-balance generators L_{ϵ_j}

remain self-adjoint with respect to the relevant inner product. By functional calculus, the spectral gap is

$$\bar{\alpha}_\epsilon = \sum_j \lambda_j \frac{\alpha}{1 + \epsilon_j \alpha},$$

and the inequality

$$\bar{\alpha}_\epsilon \geq \alpha \sum_j \lambda_j (1 - \epsilon_j \alpha).$$

C.2 Strong convergence on the generator domain

It is standard to check strong convergence in the domain and thus p -norm convergence. Indeed, p -norm convergence follows from σ -strong convergence for bounded sequences (or from strong convergence for bounded nets), as stated in the next lemma.

Lemma C.2. *Let $(\mathcal{M}, \phi)$ be a von Neumann algebra with normal faithful state ϕ and let $(x_\alpha) \subset \mathcal{M}$ be a bounded net (or sequence) converging strongly (resp., σ -strongly) to $x \in \mathcal{M}$. Then for every $1 \leq p < \infty$ and every $\eta \in [0, 1]$, in Kosaki's notation $\iota_\eta(x_\alpha) \rightarrow \iota_\eta(x)$ in $L_p(\mathcal{M}, \phi)_\eta$ -norm. Moreover, if $\iota_\eta(x_\alpha) \rightarrow \iota_\eta(x)$ in $L_p(\mathcal{M}, \phi)_\eta$ -norm for $1 \leq p < \infty$, then it is true for all $0 < p < \infty$.*

Proof. This follows directly from the Riesz-Thorin interpolation theorem, as shown in [23, Lemma 2.3]. $\square$

Proposition C.3. *Let $L \geq 0$ be a self-adjoint operator on $L^2(\mathcal{M})$, then $L_\epsilon(x) \rightarrow L(x)$ in the strong* topology for every $x \in \text{Dom}(L)$. Thus in Kosaki's notation, $\iota_\eta(L_\epsilon(x)) \rightarrow \iota_\eta(L(x))$ in $L_p(\mathcal{M}, \phi)_\eta$ -norm for $x \in \text{Dom}(L)$. Moreover, assuming that $\mathcal{A} = \text{Dom}(L) \cap \mathcal{M}$ is a weak*-dense $*$ -algebra, $\Gamma_\epsilon(x, y) \rightarrow \Gamma(x, y)$ strongly for all $x, y \in \mathcal{A}$. Similarly, in Kosaki's notation, it converges in the p -norm.*

Proof. Since $L \geq 0$ is self-adjoint, its spectral resolution satisfies

$$L = \int_0^\infty t dE_t.$$

For $x \in \text{Dom}(L)$, the functional calculus gives

$$L_\epsilon x = \int_0^\infty \frac{t}{1 + \epsilon t} dE_t x, \quad Lx = \int_0^\infty t dE_t x.$$

Hence

$$\|L_\epsilon x - Lx\|_{L^2(\mathcal{M})}^2 = \int_0^\infty \left| \frac{t}{1 + \epsilon t} - t \right|^2 d\|E_t x\|^2 = \int_0^\infty \frac{\epsilon^2 t^4}{(1 + \epsilon t)^2} d\|E_t x\|^2.$$

The integrand converges pointwise to 0 as $\epsilon \downarrow 0$, and for all $t \geq 0$, we have the bound

$$0 \leq \frac{\epsilon^2 t^4}{(1 + \epsilon t)^2} \leq t^2.$$

Since $x \in \text{Dom}(L)$ implies $\int_0^\infty t^2 d\|E_t x\|^2 < \infty$, the Dominated Convergence Theorem gives $\|L_\epsilon x - Lx\|_{L^2(\mathcal{M})} \rightarrow 0$. Moreover, $\|L_\epsilon x\|_\infty \leq \|Lx\|_\infty$, because $(I + \epsilon L)^{-1}$ is a contraction on $\mathcal{M}$. Uniform boundedness together with L^2 -convergence therefore yields strong convergence. Thus $L_\epsilon \rightarrow L$ strongly on $\text{Dom}(L)$. It follows from L_ϵ being self-adjoint that $L_\epsilon^*(x) \rightarrow L^*(x)$ for every $x \in \text{Dom}(L)$. Hence, the convergence is strong* on $\text{Dom}(L)$.

For $x, y \in \mathcal{A}$, the above convergence permits a direct comparison of $\Gamma_\epsilon(x, y)$ and $\Gamma(x, y)$.

Given $L_\epsilon(x) \rightarrow L(x)$ strongly and $L_\epsilon(x)^* \rightarrow L(x)^*$ strongly, we obtain

$$\|L_\epsilon(x)^* y - L(x)^* y\|_{L^2(\mathcal{M})} \rightarrow 0, \quad \|x^* L_\epsilon(y) - x^* L(y)\|_{L^2(\mathcal{M})} \rightarrow 0.$$

Since $\mathcal{A}$ is a $*$ -algebra, $x^* y \in \mathcal{A} \subseteq \text{Dom}(L)$, so $L(x^* y)$ is well defined. By the definition of strong* convergence,

$$\|L_\epsilon(x^* y) - L(x^* y)\|_{L^2(\mathcal{M})} \rightarrow 0.$$

All terms appearing in eq. (30) converge strongly. Hence $\Gamma_\epsilon(x, y) \rightarrow \Gamma(x, y)$ strongly for all $x, y \in \mathcal{A}$.

We also check that $\Gamma_\epsilon(x, x)$ is uniformly bounded in $\mathcal{M}$ as by [11, Prop 2.5]: $(1 + \epsilon L)^{-1}$ is bounded, completely positive normal contractions on $\mathcal{M}$, by interpolation we have $\|L_{\epsilon, p}(x)\|_{L^p(\mathcal{M})} \leq \|L_p(x)\|_{L^p(\mathcal{M})}$. For $x \in \text{Dom}(L)$ we have $L(x) \in \mathcal{M}$, and therefore each term appearing in eq. (30) is uniformly bounded in $L^{p/2}(\mathcal{M})$. By Lemma C.2, we obtain the p -norm convergence stated for $1 < p < \infty$. $\square$

Remark C.4. From L^1 convergence as in eq. (41) and uniform boundedness of $\Gamma_\epsilon(x, x)$ in $\mathcal{M}$ for $x \in \text{Dom}(L)$, we can apply the interpolation trick in [23] to check L^p norm convergence for $1 \leq p < \infty$.

D General case: noncommutative diffusion semigroup

We address the complication that $\Gamma(x, y)$ involves only weak* convergence. To obtain convergence in the L^p -norm, one cannot pass directly to the limit $\epsilon \downarrow 0$. By an argument due to the unpublished work of the first author and collaborators [27], one can form convex combinations of the regularized terms $\sum_j \lambda_j \Gamma_{\epsilon_j}(x, y)$ that converge in $\|\cdot\|_{L^1(\mathcal{M})}$ to $\Gamma(x, y)$, for $\epsilon_j > 0$, $\lambda_j \geq 0$ and $\sum_j \lambda_j = 1$. In this way, norm convergence is recovered at the expense of convexification. We begin by introducing some definitions and results from [27].

Definition D.1 (noncommutative Diffusion QMS). *A semigroup (T_t) is called a noncommutative diffusion process if for every $x \in \text{Dom}(L) \cap \mathcal{M}$ the gradient form satisfies $\Gamma(x, x) \in L^1(\mathcal{M})$.*

Proposition D.2. *(T_t) is noncommutative diffusion if*

$$\Gamma_\epsilon(x, x) \rightarrow \Gamma(x, x) \in L^1(\mathcal{M}) \quad (41)$$

in $\sigma(L_1(\mathcal{M}), \mathcal{M})$ topology. This implies $\Gamma(x, x) \in \overline{\text{conv}(\Gamma_\epsilon)}^{\|\cdot\|_{L^1(\mathcal{M})}}$ by Hahn-Banach. Thus, there exists a family of Lindbladians L_{ϵ_j} , and the gradient form of convex combinations of Lindbladians also converges

$$\Gamma_{\sum_j \lambda_j L_{\epsilon_j}}(x, y) = \sum_j \lambda_j \Gamma_{\epsilon_j}(x, y) \rightarrow \Gamma(x, y)$$

in $\|\cdot\|_{L^1(\mathcal{M})}$ as $\{\epsilon_j\} \downarrow 0$, for some $\epsilon_j > 0$, $\lambda_j \geq 0$ and $\sum_j \lambda_j = 1$.

We now complete the proof of the general case. We first convexify the regularized generators, apply the Poincaré inequality on each finite slice, and then take the limits in the order $k \rightarrow \infty$ followed by $n \rightarrow \infty$. Writing $q = p/2$, for the fixed pair (y, y) , choose finite convex combinations

$$\bar{L}_n := \sum_{j=1}^{m_n} \lambda_{n,j} \hat{L}_{\epsilon_{n,j}}, \quad C_n := \Gamma_{\bar{L}_n} = \sum_{j=1}^{m_n} \lambda_{n,j} \hat{\Gamma}_{\epsilon_{n,j}},$$

where $\lambda_{n,j} \geq 0$, $\sum_j \lambda_{n,j} = 1$, and $\delta_n := \max_j \epsilon_{n,j} \rightarrow 0$, such that

$$\left\| \iota_{\frac{1}{2}}(C_n(y, y) - \hat{\Gamma}(y, y)) \right\|_{q, \hat{\phi}, \frac{1}{2}} \rightarrow 0. \quad (42)$$

The preceding Hahn–Banach convexification gives L^1 -convergence, and the interpolation argument in Theorem C.4 gives the L^q -convergence in eq. (42). The spectral gap of $\bar{L}_n$ is

$$\bar{\alpha}_n := \sum_{j=1}^{m_n} \lambda_{n,j} \frac{\alpha}{1 + \epsilon_{n,j} \alpha},$$

and $\bar{\alpha}_n \rightarrow \alpha$ because $\delta_n \rightarrow 0$.

For each k , let $\bar{L}_{n,k} := \bar{L}_n|_{\mathcal{M}_k}$. Each regularized generator commutes with $\mathcal{E}_k$, so $\bar{L}_{n,k}$ is a τ_k -symmetric Lindbladian with spectral gap $\bar{\alpha}_n$ and the same fixed-point expectation E_k as L_k . Applying the tracial PI(p, p) inequality to $\bar{L}_{n,k}$ gives

$$\begin{aligned} A_k &:= \left\| \iota_{\frac{1}{2}}(x_k - E_k(x_k)) \right\|_{p, \phi_k, \frac{1}{2}} \\ &\leq \frac{p}{\sqrt{2\bar{\alpha}_n}} \left\| \iota_{\frac{1}{2}} \left(\Gamma_{\bar{L}_{n,k}, \frac{1}{2}}^{(p,k)}(x_k, x_k) \right) \right\|_{q, \phi_k, \frac{1}{2}}^{1/2} \end{aligned}$$

$$\begin{aligned}
&= \frac{p}{\sqrt{2\bar{\alpha}_n}} \left\| \mathcal{E}_{k,1} \left(\iota_{\frac{1}{2}}(C_n(y, y_k)) \right) \right\|_{q, \widehat{\phi}, \frac{1}{2}}^{1/2} \\
&\leq \frac{p}{\sqrt{2\bar{\alpha}_n}} \left\| \iota_{\frac{1}{2}}(C_n(y, y_k)) \right\|_{q, \widehat{\phi}, \frac{1}{2}}^{1/2}.
\end{aligned}$$

Here the equality follows from the bimodularity of $\mathcal{E}_k$ and its intertwining with each $\widehat{L}_{\epsilon_{n,j}}$, while the last inequality follows from the contractivity of its L^q extension.

For fixed n , since C_n is a finite convex combination of bounded regularized gradient forms and $y_k \rightarrow y$ σ -strongly,

$$\iota_{\frac{1}{2}}(C_n(y, y_k)) \xrightarrow{k \rightarrow \infty} \iota_{\frac{1}{2}}(C_n(y, y)) \quad \text{in } L^q(\widehat{\mathcal{M}}).$$

On the other hand, eqs. (27) and (28) gives convergence of the left-hand side of the Poincaré inequality. Therefore, taking $k \rightarrow \infty$ with n fixed yields

$$\left\| \iota_{\frac{1}{2}}(x - E(x)) \right\|_{p, \phi, \frac{1}{2}} \leq \frac{p}{\sqrt{2\bar{\alpha}_n}} \left\| \iota_{\frac{1}{2}}(C_n(y, y)) \right\|_{q, \widehat{\phi}, \frac{1}{2}}^{1/2}.$$

Finally, let $n \rightarrow \infty$. Using eq. (42) and $\bar{\alpha}_n \rightarrow \alpha$, we obtain

$$\left\| \iota_{\frac{1}{2}}(x - E(x)) \right\|_{p, \phi, \frac{1}{2}} \leq \frac{p}{\sqrt{2\alpha}} \left\| \iota_{\frac{1}{2}}(\widehat{\Gamma}(y, y)) \right\|_{q, \widehat{\phi}, \frac{1}{2}}^{1/2}.$$

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